Beaches or Books: Tourism and Human Capital Accumulation*

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Abstract

It is well established that household schooling decisions for their children are determined by a combination of forward looking expectations about the returns to education, and the opportunity costs of lost wages. What is still not well understood is the effect of positive shocks to low and mid-skilled services on a household's educational attainment decision making. I fill this gap in the literature by studying the impacts of increases in tourism intensity on schooling in the context of Jamaica, where tourism exports comprise a major share of economic output. Exploiting temporal and geographic variation in tourism activity across the country, I investigate the effects of tourism shocks on enrollment, selection into different skill levels in the labor force, attendance, and academic expenditures. Higher previous-year tourism levels reduce the probability that exposed students enroll in tertiary schooling or pass advanced pre-university examinations. Increases in tourism result in lower illness-induced absenteeism, and slightly higher transportation-cost driven absences for rural students. I consider the implications of these findings for middle-income countries that specialize in tourism or similar low and mid-skilled services, but that also want to improve educational outcomes for their populations.

1 Introduction

Human capital formation is a crucial component of structural transformation and economic growth. Growth of a country's human capital is dependent in large part on household decisions about human capital investment. It is therefore important to understand how economic factors shape investments in schooling, the manner in which the growth of specific sectors can influence these investments, and the degree of heterogeneity in observed effects across socioeconomic status, and geography. The existing literature has studied how booms in export

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manufacturing Atkin (2016), agricultural productivity shocks (Shah and Steinberg 2017), and commodity price shocks (Carrillo 2020) among others, can have substantial impacts on education attainment. However, there has been a lack of scholarly work studying the ways in which shocks to low and medium-skilled services influence human capital investment in exposed areas. This paper addresses this gap in the literature by examining the effects of specialization in tourism services on schooling enrollment and investments in Jamaica.

This paper answers the question: Do higher levels of local tourism affect human capital investment? More broadly, this study examines the impact of specialization in low and medium-skilled services on educational outcomes in a small open economy. This question is especially relevant to a sizable group of lower and middle-income countries in regions such as Sub-Saharan Africa and Latin America that are transitioning directly from economies based around natural resource extraction, to those focused on services (Nayyar et al. 2021). Theoretically, the effects of these transitions depend on two channels: an income effect, whereby higher earnings lower the marginal utility of consumption and encourage greater investment in schooling, and a substitution effect, whereby higher wages raise the opportunity cost of schooling. Influencing both of these channels are the expected returns to continuing education. Which effect dominates in cases of low and medium-skilled service sector specialization is the empirical question at the heart of this paper.

As an upper middle-income country which has invested considerably in the export-focused service of tourism, Jamaica is an excellent setting in which to study the effects of specialization in low and medium-skilled services on schooling. Like many emerging economies, over the past half-century Jamaica's economy has transitioned from being based upon natural resources (Bauxite and Agriculture), to one being based upon services such as tourism (King 2001). Tourism services comprise over 33 percent of economic activity when considering back-linkages (World Bank 2022), and earnings have grown by an average of 5 percent since 2000 (Jamaican Ministry of Tourism 2025). The country has, however, struggled to maintain consistent GDP growth [1], and an underlying cause has been the lack of growth in labor productivity.[2] Given the importance of human capital investments for labor pro-

[1]Source: (World Bank 2022)

[2]See: Caribbean National Weekly

ductivity,(Lucas 1988) it is important to understand whether growth in tourism services disincentivizes households from doing so.

My study considers three related dimensions of household human capital investment decision making: 1) enrollment at different levels of schooling; 2) attendance and factors that impede it; and 3) spending on different components of education. Successful completion of progressively higher levels of schooling grants access to occupations requiring progressively higher levels of skill. Enrollment decisions effectively act as career skill level decisions, with a significant amount of path dependence (Carrillo 2020). Given that specialization in the production of high-skilled goods and services is a necessity for reaching high-income status, these enrollment decisions in aggregate are extremely consequential for an emerging economy. Conditional on enrollment, regular attendance is known to be critical for academic mastery (Gershenson et al. 2017), and thus the effects of economic shocks on schooling consistency in addition to basic enrollment is also relevant. Finally, the level of household spending on items such as textbooks and supplies, or services like extra lessons may also impact the ability of children to fully benefit from enrollment and regular attendance. This paper shows how shocks to tourism impact these important features.

To carry out this analysis, I combine unique and granular tourist spending data with detailed schooling data from Jamaican household surveys for the years 2006, 2010, 2012, 2014, and 2018. Data on tourism activities comes from exit surveys collected by the Ministry of Tourism on a representative sample of visitors to the island. Data on household human capital investments come from the Jamaica Survey of Living Conditions (JSLC), a nationally representative household survey managed by the Statistical Institute of Jamaica (STATIN). The JSLC provides information on enrollment, absenteeism, grade level, type of school attended, and education spending at the individual level across categories such as tuition, textbooks, and transportation. I link households and one-year tourist expenditures across spatially consistent units, producing a repeated cross-section of over 13,000 school-aged children and young adults between the ages of 5 and 19. I use a shift-share instrumental variable strategy to account for the likely endogeneity of development-area tourism levels, leveraging plausibly exogenous variation in visits to Jamaica by tourists from different countries of origin.

The results show that higher area tourism levels produce no change in the likelihood that individuals aged 5 to 19 are enrolled in grade schools, nor any change in the relative popularity of different reasons provided for dropping out. Higher tourism levels do reduce the likelihood of college-aged adults being enrolled in a university of any kind. I interpret this result as suggesting that higher tourism levels raise the opportunity cost of a college education. In terms of schooling attendance, I obtain mixed results, with higher prior-year tourism levels reducing frequent and occasional absenteeism resulting from illnesses, but also slightly increasing absenteeism attributed to transportation costs. I argue that the first result indicates positive spillovers to education from higher earnings resulting from increased spending on healthcare services from tourism booms as was observed in McKetty (2026). Additionally, I argue that the increase in transportation-related absences reflects a crowding out of local consumption of transportation services by tourists, which I support by demonstrating that the effect is entirely driven by rural students, who in general must travel further for schooling.

The findings of this study are relevant to research and policy on school enrollment, the intersection of health and human capital accumulation, and the spillover effects of local sector booms on educational attainment. Unlike the boom in Mexican export manufacturing between 1986 and 2000 studied by Atkin (2016), the expansion of tourism services does not reduce completion of secondary school, but it does limit progression to tertiary schooling. This may reflect the fact that tourism is largely composed of mid-skilled occupations, or that the major beneficiaries of increases in local tourism services are workers in mid-skilled positions in sectors outside of tourism(McKetty 2026). I also highlight the likely importance of social safety net programs in helping enrollment levels to remain consistent regardless of changes in tourism earnings. These results may point towards the necessity of such programs for preventing drop out rates from increasing in response to shocks to low and medium skilled sectors such as tourism.

The finding that higher tourism is associated with lower illness-related absenteeism makes clear the importance of economic shocks not just for the choice to enroll in school, but for the ability to benefit from that enrollment. This result emphasizes a feature that distinguishes tourism from sectors such as export manufacturing and agricultural commodities: local consumption of the product. This characteristic of tourism services means that evaluation of potential spillovers to local populations must go beyond focusing on backlinkages, and must also take account of how tourists may compete with and exclude locals from important services through inflationary pressures. The potential for "crowding out" (Allen et al. 2021) spillovers from positive shocks to tourism to exacerbate inequalities in unforeseen domains such as education is an important consideration that has not been thoroughly studied in developing country contexts until now.

This paper contributes to three areas of the literature, with the first being the expansive literature studying the factors influencing human capital formation. Some work in this area has focused on the role of the expansion of export-lead industrialization on schooling decisions (Nguyen 2025; Cai et al. 2024; Atkin 2016; Heath and Mobarak 2015), with studies such as those by Bau et al. (2020), Edmonds et al. (2010), and Edmonds and Pavcnik (2005) showing the important role played by child labor. Other studies have examined regions and countries specializing in agricultural commodities, and have used shocks to international prices (Beshir and Maystadt 2024; Carrillo 2020), or to weather (Zimmermann 2020; Shah and Steinberg 2017) as identifying variation. I contribute to this broad literature by illustrating the importance of channels such as health and access to transportation services on educational outcomes. Furthermore, this paper shows how shocks to mid-skilled services can both encourage completion of secondary schooling, while also reducing investment in tertiary education, an important mechanism for developing high-skilled labor.

This study also contributes to the literature studying services-lead structural transformation in open economies. One strand of this literature has shown the capacity of high-skilled services to drive productivity growth in contexts such as India (Peters et al. 2026; Fan et al. 2023). Rodrik (2016) first coined the term "premature de-industrialization" to describe the phenomenon of many emerging economies transitioning directly from natural resources to services, and described the uncertain implications of this divergent transition path. This paper contributes by studying the impacts of the growth of a major low and medium-skilled service sector on education decisions across all levels of schooling. This paper also highlights how the local consumption aspect of tourism services can crowd out local consumption and harm access to schooling as a result.

Finally, this paper also speaks to the “resource curse" literature which investigates the extent to which natural resource abundance can harm long-run development. Seminal work by Corden and Neary (1982) described how natural resource specialization can crowd out domestic manufacturing through currency appreciation, which they dubbed the “Dutch Disease". Other research in this space has focused on topics such as resource booms and human capital accumulation (Bonilla Mejía 2020), natural resources and backlinkages (Aragón and Rud 2013), and the importance of institutions for effective use of natural resources for development (Venables 2016). Natural endowments such as weather, beaches, and topography are crucial features for the ‘Sun, Sand, and Sea’ style of tourism that is common in Jamaica and other developing countries (Faber and Gaubert 2019; King 2001). Tourism sectors in developing contexts are also criticized for acting as enclaves that do not broadly benefit local populations. This paper highlights a channel by which natural endowments monetized as a service, can impact human capital accumulation in developing country contexts.

The paper is organized as follows. Section 2 will provide an overview of the Jamaican tourism industry and education system. Section 3 provides an overview of the data used in the analysis. Section 4 discusses the empirical approach for the study. Section 5 details the results, and section 6 concludes.

2 Context and Background

2.1 Jamaica and Tourism

Jamaica is an upper-middle-income country in the northern Caribbean with an economy heavily based on services, which comprise about 70 percent of the nation’s Gross Domestic Product (GDP).[3] Tourism services account for 33 percent of GDP when factoring in backward linkages. Even as the industry has seen significant expansion over the past three decades,

[3]Source: STATIN: Annual GDP Statistics

the country has struggled to achieve meaningful economic growth. Over the past thirty years the tourism sector has expanded by 36 percent, and overall GDP by only 10 percent.[4] Low labor productivity resulting in part from a lack of high-skilled workers is one of the most often-cited reasons for this challenge.[5]

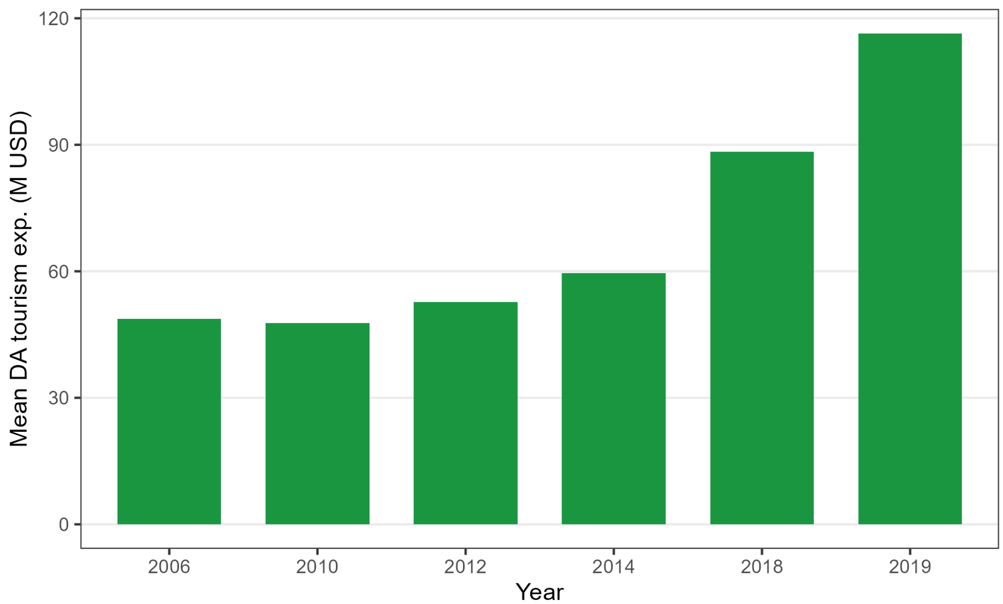

Figure 1: Mean Development-Area Tourism Expenditure by Year
Mean real accommodation expenditure (2024 USD) per development area by JSLC survey year.

2.2 The Jamaican Education System

Schooling in Jamaica is compulsory from primary school beginning in grade 1 up until grade 11.[6] The system is divided into primary and secondary stages of schooling, with primary schools teaching grades 1 through 6, and secondary schools teaching grades 7 through 13.[7] The crucial milestone in the secondary stage that helps determine college attendance is the Caribbean Secondary Education Certificate examination, taken in the 11th grade.[8] These exams cover five major subject areas and are the primary secondary credential.[9] These

[4] Source: Vision 2030 Jamaica: Jamaica's Tourism Sector is Resilient

[5] Source: (Caribbean Policy Research Institute 2023)

[6] Enrollment in early childhood education is not required.

[7] Source: (The World Bank 2021)

[8] This is based upon the British O-Level Examination. Completion of these exams is equivalent to receiving a high school diploma in the United States.

[9] Source: CSEC Equivalent in USA – American Education & Translation Services (AET)

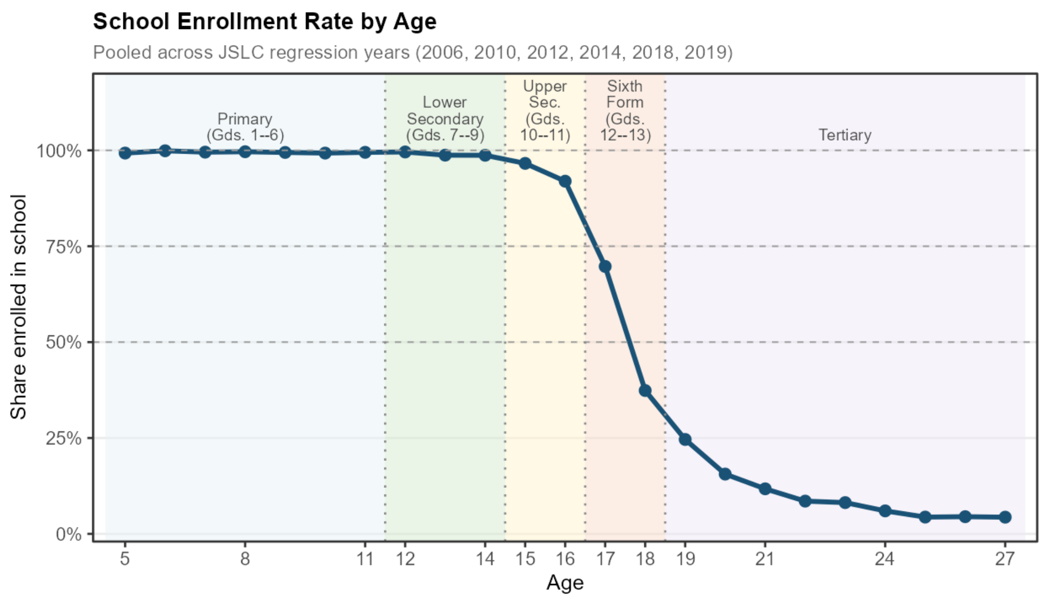

Figure 2: School Enrollment Rate by Age
Share of individuals enrolled in school by single year of age, pooled across JSLC regression years (2006, 2010, 2012, 2014, 2018, 2019). Shaded bands denote Jamaican school stages; dashed reference lines mark 50%, 75%, and 100% enrollment.

examinations provide entry to skilled work, and are a necessary pre-requisite for those who want to continue on to university.

Successful completion of the CSEC exams grants access to Sixth Form: two years of additional study in three or four subject areas designed to prepare students for university.[10] At the end of Sixth Form (grade 13), students sit the Caribbean Advanced Proficiency Exam (CAPE).[11] Jamaica has nearly universal enrollment in schools through grade 11, with 96.9 percent of 15 and 16 year olds enrolled in Upper Secondary School (which covers grades 10-11) in 2019.[12] There is a significant drop-off in enrollment for non-compulsory Sixth Form coursework, and even larger drop in the enrollment rate for tertiary schooling as shown in figure 2. In 2019, the last year covered by my analysis, the share of 17-19 year olds in Sixth Form was 45 percent, and the share of 20-22 year olds in post-secondary schooling was 12

[10] These courses are roughly equivalent to Advanced Placement or International Baccalaureate classes in the U.S. context

[11] Source: American Education and Translation Services: Credential Evaluation in USA

[12] See: Table 46

percent.

Primary and secondary schooling is largely administered by the government and nominally free, while early childhood education is primarily provided by private institutions with some government support. Significant costs remain for families, particularly at the secondary level, including uniforms, textbooks, transportation, and meals. Households spend the equivalent of 3 percent of GDP on early childhood, primary, and secondary education. Relative to the average GDP per capita in 2017 of approximately 3,053 USD, the household cost for one child represented 13 percent of GDP per capita for primary school and 20 percent for secondary education.[13] Financial constraints are unsurprisingly the most commonly cited reason for leaving school before grade 11, as shown in Table **??**, with lunch, snacks, and transportation comprising the largest reported shares of schooling expenditures.

The Jamaican government has made considerable investments in the education system over the previous two decades. These have included a conditional cash transfer (CCT) program to encourage attendance[14], Ministry of Education-funded lunch and snack programs[15], and support for low-income families to cover textbooks and other materials.[16] Public expenditure on education as a share of GDP was approximately 5 percent throughout the 2010s, similar to the average for industrialized countries in the Organization for Economic Cooperation and Development (OECD) (Chegwin et al. 2021).

There is near-universal school attendance up to age 16 (World Bank 2022), but Jamaican students perform worse than those in peer nations on measures of academic achievement.[17] Jamaican students complete an average of 11.4 years of formal education, but only 7.1 Learning-Adjusted Years of Schooling (LAYS) (World Bank 2022), reflecting significant

[13] Source: (The World Bank 2021)

[14] The Programme of Advancement through Health and Education (PATH) is a CCT program overseen by the government, and implemented nationwide in 2002. Qualifying households receive bimonthly transfers that are conditioned on school attendance and that vary based on child gender and grade level (Stampini et al. 2018; Levy and Ohls 2010).

[15] Jamaica has a National School Feeding Program that provides lunch and snacks to vulnerable students. Source: Jamaica Invests in the Future

[16] Sources: 1. Education Ministry Providing Textbooks,2. Parents Urged to Take Advantage of Book Rental Scheme

[17] Source: Jamaica Education Report Card 2025: Assessing the Present, Guiding the Future (Caribbean Policy Research Institute et al. 2025).

gaps in schooling quality. Another key challenge is secondary school completion and tertiary enrollment: approximately 90 percent of students complete secondary education, while only 27 percent enroll in tertiary institutions. The transition from secondary to tertiary education is therefore one of the most important margins to examine.

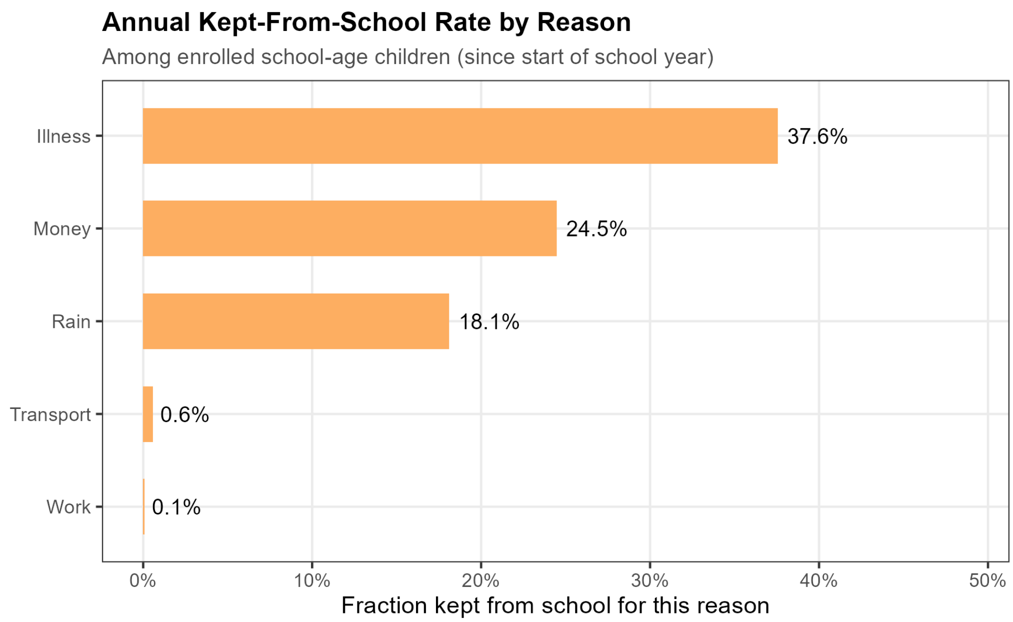

Figure 3: Annual Kept-From-School Rate by Reason
Share of enrolled school-age children kept from school for each reason at least once since the start of the school year, pooled across JSLC waves.

2.3 Hypotheses Regarding Tourism and Human Capital Investments

The literature on human capital investment stresses that optimal schooling decisions by households are shaped by a combination of the perceived returns to additional schooling in the form of more lucrative career outcomes, the opportunity costs of continuing education (lost wages), the economic cost of education (school fees, expenditures on supplies, etc.) (Carrillo 2020; Shah and Steinberg 2017; Atkin 2016). Thus, in this context, the effect of a positive tourism shock on schooling enrollment and investment will depend on the extent to which the shock alters the net returns to schooling, and raises household consumption and thus future human capital. The perceived net returns to additional education depend on the costs of obtaining said schooling, and whether the sectoral boom benefits occupations and career paths that require further academic qualifications.

If tourism industry growth generates higher wages or employment opportunities for jobs demanding a college degree, and the costs of obtaining said degree are not prohibitive, then we would expect more forward-looking households to increase level of schooling attained by their children. However, if an expanding tourism sector increases wages and employment in jobs only requiring through grade 11, and there is no change in the returns to high-skilled/high-education employment, it is likely that there will be a reduction in schooling attainment for some households. The insights from the literature on the competing determinants of household human capital attainment inform two hypotheses: If there is an increase in development area tourism levels

1. If the substitution effect dominates, an increase in tourism levels will decrease enrollment and attendance in the upper levels of secondary school, and in universities since the sector is largely composed of mid-skilled labor
2. If the income effect dominates, higher tourism levels will increase enrollment and attendance in upper levels of secondary school and universities owing to higher earnings and consumption.

The first hypothesis reflects the finding in McKetty (2026) that increases in real expenditures resulting from higher tourism levels are approximately 1.8 percent per year and are heavily concentrated in food and healthcare. The utility gained from greater consumption and improved health may be perceived as more valuable than additional investment in schooling, particularly given the substantial jump in expenses necessary for tertiary education in Jamaica (The World Bank 2021). The second hypothesis reflects the fact that lack of money is one of the main drivers of school absences.[18]

[18] Source: Figure 3

3 Data

Data on tourist expenditures come from exit surveys conducted by the Ministry of Tourism on a sample of departing tourists leaving through one of the island's two international airports. Enumerators randomly select traveling parties to survey in the departure halls. Because all stopover tourists (visitors who spend more than 24 hours in Jamaica) must depart through these airports, the surveys obtain a representative sample of non-cruise visitors. Respondents answer detailed questions about their spending, demographics, and, critically, the hotel at which they stayed. This provides a representative picture of the distribution of tourist spending across Jamaican locales. I scale the survey estimates using aggregate arrival and spending data from annual reports published by the Ministry of Tourism. Tourism data span the years 2004–2006 and 2008–2019, covering approximately 50,000 traveling parties and 100,000 individuals.

Education data come from the Jamaica Survey of Living Conditions, a nationally representative household survey conducted annually by the Statistical Institute of Jamaica. The JSLC collects questions at both the household and individual levels. In the education module, questions on enrollment are asked for all individuals. For currently enrolled children and youth, detailed questions cover the type of school attended, academic progress and exam performance, and spending by parents across categories such as tuition, textbooks, and transportation. Individuals no longer in school answer questions about their educational attainment, skill levels, and, for those who dropped out before grade 11, their reasons for doing so. The JSLC years used in this paper are 2006, 2010, 2012, 2014, 2018, and 2019, covering 8,941 households and 12846 children and young adults between the ages of 5 and 18. The college-aged sample including individuals aged 19-27 has 6312 persons. The full breakdown can be seen in table 1. Administrative shapefiles from STATIN enable me to link tourism spending and households across spatially consistent geographic units.

Table 1: Sample Overview

Panel A: Sample characteristics	
Total person-year observations	19,158
Household-years	8,941
Development areas	57
Survey years	2006, 2010, 2012, 2014, 2018, 2019
Age group composition (% of sample)	
Primary (ages 5–11)	6,224 (32.5%)
Lower secondary (ages 12–14)	2,832 (14.8%)
Upper secondary (ages 15–16)	1,933 (10.1%)
Sixth form (ages 17–18)	1,857 (9.7%)
College-age (ages 19–27)	6,312 (32.9%)
Female	49.8%
Urban	38.3%
Rural	61.7%
Panel B: School enrollment (ages 5–19, regression years)	
Overall enrollment rate	87.9%
Primary age (5–11)	99.5%
Lower secondary (12–14)	99.0%
Upper secondary (15–16)	94.4%
Sixth form age (17–19)	44.9%

Notes: Full regression sample, ages 5–27, pooled across JSLC years 2006, 2010, 2012, 2014, 2018, and 2019. Primary = ages 5–11; Lower Secondary = ages 12–14; Upper Secondary = ages 15–16; Sixth Form = ages 17–18; Tertiary-age = ages 19–27. Urban/rural classification follows STATIN development-area designation. Poverty indicator based on the JSLC consumption poverty measure.

4 Empirical Approach

I calculate tourism at the development-area level because development areas are government-designated districts with similar economic and demographic characteristics. These areas provide a natural grouping for communities and economic activities, and include both rural and urban components, allowing for estimation of heterogeneous rural-urban impacts. The main empirical specification is the following,

$$Y_{idt} = \alpha + \beta\, \widehat{\text{Tourism}}_{dt} + \mathbf{X}'_{idt}\gamma + \delta_d + \lambda_t + \varepsilon_{idt} \qquad (1)$$

where Y_{idt} is an education outcome for individual i in development area d in year t; $\widehat{\text{Tourism}}_{dt}$ is the fitted value of local tourism accommodation expenditures from the first stage using B_{dt} as the excluded instrument; $\mathbf{X}_{idt}$ is a vector of individual and household controls; δ_d and λ_t are development area and year fixed effects; and ε_{idt} is a disturbance term clustered at the development area level. The vector of controls contains an indicator for whether the student is female, their age, and the number of siblings aged 19 years and younger who are enrolled in school.

I use a one-year lag of tourism revenue, reflecting the empirical finding in the literature that schooling decisions are influenced by conditions in prior periods. I do not include additional lags because of likely serial correlation between immediately adjacent years. I run robustness checks using contemporaneous and two-year lags.

4.1 The Shift-Share Instrumental Variable Approach

The central empirical challenge is that the level of tourism activity in a development area may be correlated with local economic conditions that independently affect household education decisions. More touristic development areas may, for example, have systematically better governance, resulting in both higher-quality schools and greater hotel investment. Such endogeneity would bias OLS estimates. To address this, I instrument area revenues using a shift-share instrumental variable. I exploit year-over-year variation in the intensity of tourist arrivals from different regions and countries, interacted with the heterogeneous preferences of tourists from different origin countries for different parts of Jamaica.

The instrument is constructed by interacting aggregate "shifts" - — with predetermined "shares". The Bartik instrument is:

$$B_{dt} = \sum_{o} s_{od} \cdot \Delta N_{ot} \tag{2}$$

The shift-share IV is constructed by interacting an exposure "share" (capturing each development area d's historical exposure to tourists from origin country o, s_{od}) component with aggregate "shifts" (year-to-year changes in the number of tourists from each origin country o arriving in Jamaica, ΔN_{ot}). The exposure share s_{od} for development area d is the share of its tourism revenues in a base year t_0 earned from tourists originating from country o. The shift ΔN_{ot} measures the percentage change in spending by tourists from region o across all of Jamaica (excluding area d) between year $t-1$ and the base year t_0.

Exogeneity can be established either by arguing that the base-year exposure shares are orthogonal to the outcome or that the shocks themselves are orthogonal. This paper uses the shifts-based identification approach, which is the most appropriate given the context of tourist decision-making. Decisions by tourists to visit Jamaica are influenced by home-country economic conditions, conditions in competing destinations, and other factors that, conditional on year and area fixed effects, are plausibly exogenous to the development-area characteristics that influence educational investment. I estimate the effect of tourism on schooling using two-stage least squares, with the two stages of the specification given by:

$$\text{Tourism}_{dt} = \pi_0 + \pi_1 B_{dt} + \mathbf{X}'_{idt}\phi + \delta_d + \lambda_t + \nu_{dt} \tag{3}$$

$$Y_{idt} = \alpha + \beta\,\widehat{\text{Tourism}}_{dt} + \mathbf{X}'_{idt}\gamma + \delta_d + \lambda_t + \varepsilon_{idt} \tag{4}$$

where Tourism_{dt} is the observed level of tourist accommodation expenditures in development area d in year t; π_1 is the first-stage coefficient on the Bartik instrument; ν_{dt} is the first-stage error term; and all remaining notation follows equation (1).

5 Results

I first present findings on the effects of tourism levels on enrollment and sorting across career skill-tracks. I then discuss attendance and absenteeism outcomes, and conclude with a discussion of academic expenditure results.

5.1 Baseline Findings: Enrollment, Career Skill-Level Sorting, and Dropping Out

I first consider the effects of local tourism on the probability that students are enrolled in school, with a particular focus on enrollment in either university or Sixth Form. The results are provided in Table 2. First stage F-Statistics are strong across all three sample specifications.

I find no effect of prior-year development-area tourism expenditures on the likelihood that a child or young adult is enrolled in grade school, along with no effect when specifically focusing on Sixth Form enrollment in columns 1 and 2 respectively. I do find that higher tourism levels result in a statistically significant decrease in the likelihood that college-aged students are enrolled in university or have a degree the following year. The effect is significant at the 5 percent level, and the coefficient on previous-year tourism measured in hundreds of million is -0.063. The median variation among touristic development areas over five years is 31 million (McKetty 2026), suggesting that over the same period the median tourism-specializing development area experiences a 2 percentage point decrease in college enrollment relative to districts not experiencing increasing tourism levels. Since the share of individuals enrolled in university is 7.8 percent, this amounts to a 25 percent decrease in the likelihood that college-aged students in the median tourism-focused area pursue or graduate from a four year college.

Though it is not possible with the available data to determine the exact mechanism underlying the observed effect, I can still evaluate the extent to which this aligns with my hypotheses and existing theory. The decrease in the likelihood of college enrollment or attainment associated with an increase in area tourism revenues is consistent with households at least temporarily substituting away from high-skilled career investments. This would also align with the fact that job opportunities in tourism are known to be overwhelmingly mid-skilled (McKetty 2026), which typically only require completion of high school (International Labour Office 2012). Additionally, the lack of an effect of tourism earnings on grade-school enrollment is reasonable since the beneficiaries of tourism increases are largely those in mid-skilled occupations. The returns to schooling channel could, thus, potentially motivate completion of high school but not the pursuit of additional tertiary schooling.

Table 2: School Enrollment, IV Estimates

Dependent Variables: Model:	Enrolled in school (1)	Currently in sixth form (2)	In college or has degree (3)
Variables			
Tourism exp.,	-0.007	0.034	-0.063**
lag 1 (100M USD)	(0.036)	(0.034)	(0.031)
Female	0.006*	0.007	0.049***
	(0.003)	(0.006)	(0.007)
Age	-0.038***	0.004*	-0.003***
	(0.001)	(0.002)	(0.0009)
No. enrolled siblings (HH)	-0.0002	-0.002	-0.009***
	(0.002)	(0.003)	(0.003)
Fixed-effects			
Development area	Yes	Yes	Yes
Year	Yes	Yes	Yes
Fit statistics			
Bartik F-stat (1st stage)	1,030.2	446.07	567.73
Observations	13,518	5,494	5,985
R^2	0.25524	0.16714	0.04323
Sample			
School-age (5–19)	Yes	No	No
Upper secondary (14–19)	No	Yes	No
College-age (17–27)	No	No	Yes
Mean of dep. var.	0.879	0.062	0.078

Notes: Sample: children ages 5–19. All specifications include development area and year fixed effects; standard errors clustered by development area.

To further evaluate the plausibility of a ‘secondary education trap’, in table 3 I regress tourism levels on secondary school exams at the beginning, middle, and end of the secondary academic progression. The first exam in column 1, is the Grade Nine Achievement Test (GNAT), which was used to determine whether a student could progress to upper secondary school. It was phased out in 2021. Columns 2-4 are the mid-high school career examinations (Caribbean Examinations Council (CXC)) that determine readiness for advanced, pre-university coursework. Finally, column 5 denotes completion of the advanced pre-university exams necessary for admission that are conducted during Sixth Form (Caribbean Advanced

Proficiency Exams (CAPE)). If tourism is causing households to select away from high-skilled career trajectories, it is possible that this effect may also appear in performance on exams dedicated to completion of high school vs preparation for university.

The coefficient on prior-year tourism is significant only for the advanced pre-college examinations, and it is negative, equaling -0.034. For tourism-producing development areas, since the median 5 year change in tourism earnings is 31.04 million USD, this results in an average decrease of 1.1 percentage points in the share of secondary students passing the advanced pre-college examinations, which would indicate lower levels of college-readiness and a lower likelihood of acceptance to tertiary institutions. These findings lend support to the interpretation of table 2's results that tourism booms impacted educational decisions at the margin between secondary school and higher level education. This margin is also an important career track decider, as it is the difference between access to only lower and mid-skilled occupations, versus high-skilled ones.

Table 3: Examination Attainment, IV Estimates

	Lower Secondary Exit Exams	*General High School Exit Exams*			*Advanced Pre-College Exams*
Dependent Variables:	GNAT	CXC Basic	CXC General	CXC: math and English	CAPE/ A-Level
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp., lag 1 (100M USD)	-0.007	-0.019	-0.007	0.005	-0.034**
	(0.016)	(0.012)	(0.013)	(0.015)	(0.014)
Fit statistics					
Bartik F-stat (1st stage)	1,030.2	1,030.2	1,030.2	1,030.2	484.43
Observations	13,518	13,518	13,518	13,518	6,019
R^2	0.14591	0.13809	0.11882	0.05256	0.02929
Sample					
School-age (5–19)	Yes	Yes	Yes	Yes	No
A-Level age (14–20)	No	No	No	No	Yes
Mean of dep. var.	0.061	0.056	0.046	0.020	0.017

Notes: Sample varies by outcome: ages 5–19 for GNAT through CXC qualifications; ages 14–20 for CAPE/A-Level. Each outcome is a binary indicator equal to 1 if the respondent has attained at least that qualification level (or higher). Controls for gender and age included in all specifications but not shown. All specifications include development area and year fixed effects; standard errors clustered by development area.

5.2 School Attendance and Tourism Earnings

In this section I present findings for how shocks to tourism affect school attendance. In table 4 I again use binary outcome variables, with these outcomes denoting both a reason for a child's absence from school during the current academic year, and how frequently a student has missed school for said reason. In the first column, I regress an indicator for whether or not a child has been kept out of school for any reason on tourism expenditures. The coefficient on prior-year area tourism earnings is negative and statistically significant at the 5 percent level. The coefficient of -0.176 indicates that an increase in development area tourism revenues of 100 million U.S. Dollars, produces a 17.6 percentage point decline in the probability that a student misses school for any reason in an academic year. For tourism-specializing development areas, the median five year change in tourism earnings of 31.06 million would thus have resulted in a median change of -5.27 percentage points, a 28 percent decrease in annual absences based on the mean absence probability of 18.8 percent. The remaining columns investigate effects on specific types of absences.

Column 2 of table 4 shows that there is no statistically significant effect of area tourism earning on whether a student is ever kept from school for money reasons. This result is notable because 26 percent of students in the sample have been kept from school at least once because of money. In light of findings in McKetty (2026) showing that contemporaneous tourism earnings result in higher real per-capita consumption, column 2 casts doubt on the strength of the income-effect channel. Column 3 similarly shows that there is no statistically significant effect of tourism shocks on work-related absences, which are quite rare to begin with. Column 4 shows that the observed effect in column 1 is primarily driven by a decrease in frequent or occasional absences due to illness. The coefficient of -0.165 implies that for the median tourism-focused development area, growth of municipality tourism earnings results in a 5.12 percentage point decrease in the probability that students are frequently or occasionally kept from school because of illnesses. This amounts to an 81 percent decrease in frequent or occasional illness induced absences.

Considering the well-established link between school attendance and academic outcomes (Aucejo and Romano 2016), and the existing low "Learning Adjusted Years of Schooling" for Jamaican students, this result highlights an important channel through which learning capacity can be improved. As figure 3 shows, illness is the most common reason that Jamaican students are absent, making up 37.6 percent of absences. McKetty (2026) showed that around a third of the increase in real non-food expenditures for households benefiting from a boom in tourism is spent on medical services. This paper's finding that absences due to illness decrease as a result of tourism earnings further corroborates the McKetty (2026) results, and demonstrates an important spillover to human capital production not discussed by the original paper. The income effect considered in the hypotheses is not the relevant channel here. Rather this finding highlights how increased incomes allow for consumption of goods and services that are likely to improve a students ability to learn through fewer absences.

Table 4: School Absence Reasons, IV Estimates

Dependent Variables: Model:	Kept out, any reason (1)	Kept out, money (ever) (2)	Kept out, work (3)	Kept out, illness (4)	Kept out, transport (5)
Variables					
Tourism exp., lag 1 (100M USD)	-0.176** (0.086)	0.075 (0.109)	0.002 (0.002)	-0.165** (0.079)	0.020*** (0.006)
Fit statistics					
Bartik F-stat (1st stage)	863.27	863.27	863.27	863.27	863.27
Observations	11,881	11,881	11,881	11,881	11,881
R^2	0.11458	0.05459	0.00628	0.02904	0.00211
Mean of dep. var.	0.188	0.259	0.001	0.063	0.003

Notes: Sample: enrolled children; absence measured by whether a child has ever been absent for a particular reason during the school year or if they have been absent occasionally or frequently. The columns without "ever" in parentheses are for specification where the indicator is 1 if absences for the given reason has occurred either frequently or occasionally. Controls for gender, age, and enrolled household siblings are included in all specifications but not shown. All specifications include development area and year fixed effects; standard errors clustered by development area.

Finally for table 4, column 5 shows the interesting finding that tourism earnings result in a small but precisely estimated increase in the probability of frequent or occasional transportation-cost induced absences. An increase in area tourism earnings of 100 million U.S. Dollars results in a 2 percent increase in absences due to high transportation costs. The effect is far smaller in magnitude than the coefficients for illness-driven absences, and the share of students kept from school for transportation-cost-related issues at any frequency is only .6 percent[19], and .3 percent for occasional or more absences. Even so, this result is in-

[19]Source: Figure 3

teresting since McKetty (2026) found no evidence in the Jamaican context of the significant inflationary effects and crowding out of local consumption seen in contexts such as Barcelona (Allen et al. 2021). It is reasonable though, to expect that competition with tourists for local transportation services could raise prices for specific segments of the population, such as those living further away from schools. In figure 4, I disaggregate the column 5 regression by urban and rural household location status[20], and I find that rural students drive the observed increase in transit-cost related absences.

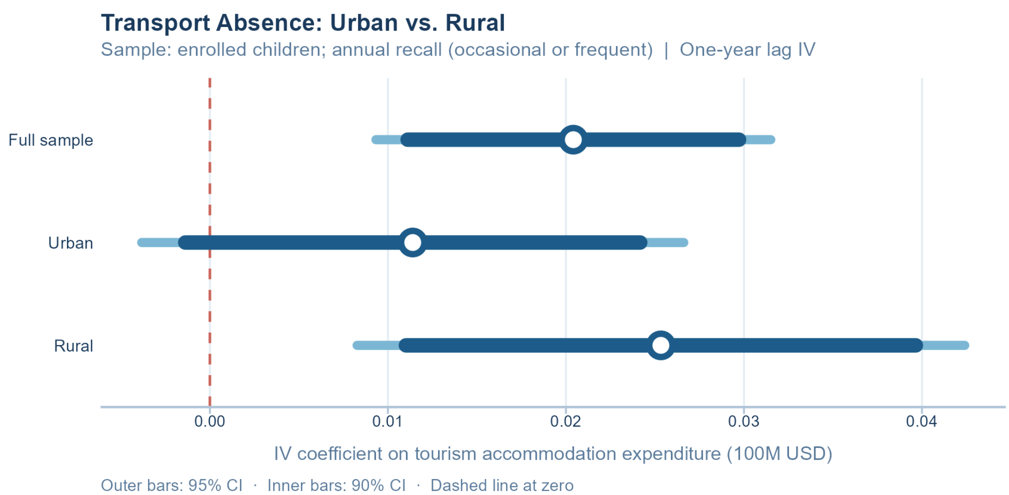

Figure 4: Transport Absence, Urban vs. Rural: IV Coefficient Estimates
Tourism expenditure coefficients (one-year lag) from the IV regressions in Table 20. Wide bars show 95% confidence intervals; narrow bars show 90% intervals. Dashed line marks zero.

Table 44 in appendix 7.8 shows that rural students live on average 1.75 kilometers farther from their schools than their urban counterparts, with rural secondary students traveling an average of 7.5 kilometers to reach their nearest secondary school compared to 3.014 kilometers on average for urban students. A similar share of both groups also rely on public transportation[21], and thus it makes sense that increased tourism activity may result in upward pressure on prices that is more burdensome for students who live further away. This result points to the potential for spillovers from tourism services, and service sector industries more broadly, to exacerbate existing inequalities even in the area of human capital accumulation. Services

[20]Development areas can contain both urban and rural segments as is shown in McKetty (2026).
[21]63.7 percent of urban students use public transit compared to 67.9 percent of rural students. See Table 45.

are defined by the local consumption of goods, whether they are traded services like international tourism, or local-facing nontradables. This local consumption is a distinguishing feature of tourism services by comparison to say the export-focused manufacturing growth of Mexico in the case of Atkin (2016), or the trade in agricultural commodities like cocoa, as in Carrillo (2020).

The transportation cost result expands beyond the income and substitution-effect focused hypotheses of earlier, and instead casts light on how positive shocks to local traded industries can impact human capital accumulation through creating impediments to easily accessing education. It is worth re-emphasizing that the small increase in absences due to transportation costs is dwarfed by the reduction in absences because of illness. Tourism booms were thus a net positive in terms of academic participation and engagement, proxied by attendance, the focus of this section of results.

5.3 School Expenditures

In the final results subsection I examine the effects of higher levels of tourism on the financial resources allocated to the schooling of children and young adults. This section enables me to capture if a shock to prior-year local tourism earnings translates into an overall change in the child-level education spending, a change in the composition of the academic budget, both or neither. In table 5 I first regress total spending on an individual's schooling on prior-year tourism levels, and then consider key components of the schooling budget. Lagged tourism levels have no statistically significant impact on the total amount a household allocates to the education of an individual child. I find the same result for each spending category with the exception of textbooks, for which the coefficient is surprisingly negative and statistically significant at the 5 percent level. An increase in area tourism spending of 100 million U.S. Dollars produces a decrease in child-level spending on textbooks of 24.5 U.S. Dollars for an academic year, a 35 percent decrease.

I show in appendix 7.4 that the effects are concentrated among upper-primary (grades 4-6) and lower-secondary (grades 7-9) school students. Table 21 reveals that there is no change in the probability that a student has the required textbooks. Focusing only on students

Table 5: School Expenditure by Category, IV Estimates

Dependent Variables:	Total spending (USD)	Tuition (USD)	Books (USD)	Transport (USD)	Uniform (USD)	Lunch and snacks (USD)
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Tourism exp.,	-14.2	-20.6	-24.5**	59.8	-53.5	48.8
lag 1 (100M USD)	(121.1)	(33.9)	(9.83)	(65.5)	(32.9)	(79.2)
Fit statistics						
Bartik F-stat (1st stage)	633.32	431.64	634.72	638.01	657.07	632.38
Observations	10,352	9,246	10,281	10,283	10,304	10,337
R^2	0.16206	0.04607	0.08583	0.07780	0.03300	0.12224
Mean of dep. var.	1037.349	48.388	70.385	258.767	88.556	464.417

Notes: Sample: enrolled children; real 2024 USD; 2014 excluded due to survey design. Total excludes tuition-including-books to avoid double-counting with the books category. Controls for gender, age, and enrolled household siblings are included in all specifications but not shown. All specifications include development area and year fixed effects; standard errors clustered by development area.

who do not have all their books, table 22 shows that higher tourism levels increase the probability that textbook prices are preventing students from having all the books needed for their courses. In table 14, I show that the coefficient estimate is capturing an extensive margin shift in book expenditures, with higher tourism resulting in some households reducing their textbook expenditures to zero.

A regression of textbook expenditures on tourism levels disaggregated by an indicator of household socioeconomic status can provide useful hints about the effects that we are observing. The challenge is that variables like poverty status, expenditure decile, and social safety net eligibility are themselves endogenous to development area tourism earnings. I instead filter the regression by the number of children or young adults belonging to a household aged 19 years or younger. I show in appendix table 23 that the number of dependents in a household is strongly correlated with socioeconomic status but is not linked to tourism levels. The coefficient estimates of the disaggregated regression can be seen in table 6 and figure 8, and the estimate for households with fewer than 3 children is negative and statistically significant. This provides suggestive evidence that relatively more affluent households are the ones reducing expenditure on textbooks to zero.

Table 6: Book Spending by Household Dependent Count, IV Estimates

	N. Under 19 (household dependents aged ≤19)	
Dependent Variable:	Books (USD)	
Model:	(1)	(2)
Variables		
Tourism exp., lag 1 (100M USD)	-16.1*	13.3
	(9.18)	(64.5)
Fit statistics		
Bartik F-stat (1st stage)	758.93	29.306
Observations	6,020	4,261
R^2	0.09269	0.06898
Mean of dep. var.	80.073	56.697

Notes: Sample: enrolled children; real 2024 USD; 2014 excluded. Column (1) restricts to households with fewer than three members aged 19 or under; column (2) restricts to households with three or more. The number of household dependents is strongly correlated with socioeconomic status (see appendix Table 23) but is not predicted by tourism levels. Controls for gender, age, and enrolled household siblings included but not shown. All specifications include development area and year fixed effects; standard errors clustered by development area.

To what can we attribute the reduction to 0 USD of spending on textbooks by more affluent Jamaican households in response to increases in tourism? One possibility is that these households are substituting towards digital versions of the textbooks that can be obtained more cheaply online. The Ministry of Education has encouraged families who are able to obtain required textbooks from independent online distributors or social media.[22]. The government has heavily invested in providing "E-Textbooks", noting that they are cheaper than paper-books. However, use of these books requires that students own a laptop computer and can regularly access the internet. Access to such technology correlates heavily with socioeconomic status,[23] and so the most privileged may be the main beneficiaries of expanded

[22]Source: Jamaica Information Service: Parents Encouraged to Take Advantage of Options in Purchasing Textbooks

[23]See: Vision 2030 Jamaica: Computer And Internet Access Among Students At The Primary And Secondary Levels

digital offerings. The fact that there is no change in the share of students able to obtain all required texts, the fact that relatively more affluent households are the ones eliminating their textbook spending, and the large disparities in access to technology services in Jamaica, all lend support to the interpretation that a subset of households are substituting from paper textbooks to digital ones.

Increases in prior-year tourism levels thus have largely no effect on the level or composition of household education spending. Indeed, there are no observed intensive margin adjustments in schooling investments. One interpretation of this finding is that initiatives such as PATH, the National Feeding Program, and the Textbook Rental Scheme help support stable human capital investments in the face of economic shocks. As a result the modest year-to-year variation in tourism earnings does not result in significant changes in households grade school spending. The statistically significant and negative effect of tourism earnings on textbook spending is the outlier among the expenditure results. While it is not possible to precisely determine the reason for this effect, the available evidence raises the possibility that the estimated coefficient is capturing substitution between different means of obtaining textbooks by more affluent households.

6 Discussion and Conclusion

This paper has studied the relationship between tourism-sector specialization and human capital investments by households in Jamaica. I combined granular surveys of tourist spending with household survey data covering absenteeism, categorized schooling expenditures, and academic performance during six years in the period between 2006 and 2019. Using an instrumental variable approach to account for the likely endogeneity of development area tourism levels, I evaluated the effect of area tourism earnings on enrollment and selection into different career skill-level tracks, attendance, and financial investments by households into each child's education. The temporal coverage of the data makes it possible to analyze the effects of tourism growth on human capital formation across a period with substantial variation in economic conditions and tourism levels across and within years. Student-level data from households covering all stages of schooling allow for analysis of the effects of tourism shocks at critical junctures for households' schooling and, by extension, career decisions.

I find that tourism levels only impact enrollment at the college or tertiary level, with the median 5-year change in tourism levels resulting in a 25 percent decline in the probability that a college-aged individual is enrolled at a tertiary institution. Median tourism growth also produces a 67 percent decrease in the probability that secondary school students have taken and passed crucial pre-college examinations. I interpret these findings as demonstrating that booms in local tourism induce households to substitute away from investing in tertiary schooling, which is consistent with tourism growth generating opportunities in occupations that primarily require a secondary level of schooling. Indeed, the lack of an effect on enrollment even for the pre-University "Sixth Form" stage of schooling contrasts markedly with the case of export manufacturing booms in Mexico, which, as Atkin (2016) showed, resulted in high school students dropping out to join the workforce.

Positive shocks to local tourism services result in a significant reduction in frequent or occasional illness-related absences, with median five-year tourism growth producing an 81 percent reduction in the probability of frequent absences. We know from McKetty (2026) that locals spend a large share of their increased earnings from tourism on medical services, which further corroborates the finding. This effect is most closely represented by what Shah and Steinberg (2017) refer to as the effect of consumption on the net impact of schooling. Regular attendance is important for academic performance (Gershenson et al. 2017), and this result demonstrates how higher earnings can contribute to more human capital accumulation at the intensive margin by reducing illnesses and the resulting disruptions to learning. I also find that tourism sector expansion increases transportation-cost induced absences for rural students who typically live farther away from their schools than urban students. Though the effect is far smaller in magnitude than the illness-related absence outcome, this brings attention to the unique potential for locally consumed service sector specialization to crowd out local consumption through inflationary effects and worsen disparities.

Finally, I show that there are few impacts of area tourism revenues on child-level household spending. There are no observed effects of tourism earnings on the total amount spent for each student in a household, and this is true across all spending sub-categories with the exception of textbooks. There is a negative effect of tourism revenues on textbook spending, which I show is a result of extensive margin changes, with a subset of the households shifting from positive book expenditures to zero such spending. Disaggregating by the number of children in a household as a proxy for socioeconomic status, I find suggestive evidence that smaller families who are on average more affluent are driving the observed effects. I argue that these findings hint towards more affluent households substituting between modes of how they obtain textbooks. The fact that the share of students having possession of all necessary books is unchanging in tourism levels, and that the study period saw increasing usage of technology in the Jamaican education system, lend support to this view.

What do these outcomes imply for Jamaica and other middle-income countries that aim to use tourism-specialization for economic development, but that also need to increase the skills of their labor forces through tertiary training to support long term growth? The results show that specialization in low to medium skilled tourism services has the potential to contribute to a "secondary education trap" where growth of the sector contributes to improved learning through reductions in absenteeism due to illness, but disincentivizes households from investing in the pursuit of tertiary education. The fact that grade school enrollment levels appear to be unrelated to variations in tourism despite the importance of the industry to the economy reflects the effectiveness of the sweeping social protection programs meant to ensure universal enrollment through grade 11(Caribbean Policy Research Institute et al. 2025).

The relative lack of robust support for tertiary schooling compared to grade school education(The World Bank 2021) likely contributes to the observed effects. As tourism growth primarily benefits occupations that do not require a tertiary degree, and because post-secondary schooling represents a burdensome investment for most Jamaican families, it is consistent with theory that the substitution effect will dominate at the college margin. The increased earnings from tourism booms are simply not large enough to make college enrollment optimal for a subset of the population. If the goal is to use tradable services made up largely of low and mid-skilled workers as a springboard to eventual specialization in high-skilled industries, interventions that lower the costs of tertiary education for households are necessary to ensure that growth of lower-skilled opportunities do not reduce educational attainment.

The finding that tourism is associated with a small but statistically significant increase in transportation-related absences among rural students points to an important mechanism by which tourism specialization (and service-sector specialization more broadly) can exacerbate existing inequalities. The crowding out of locals from transportation services because of inflationary pressure from tourism is consistent with prior work on the industry and provides evidence of a channel through which positive shocks to local tradable services can produce impediments to regular access to schools for some segments of a population. Beyond education, this result speaks to the importance of the local consumption aspect of service sector industries when evaluating their impacts on local populations. The potential for specialization in services to harm some residents through inflationary impacts must be taken into account in a way that is not often necessary for sectors such as export-oriented manufacturing. Such effects could have broad implications for local welfare and economic development in ways that are not immediately apparent, as the transportation-induced absences result makes clear.

The growth of human capital in middle-income countries depends heavily on micro-level household decisions about how much education to obtain and how much to invest at each stage. Tourism services are of growing importance in these economies, but we know little about how the growth of this industry alters household incentives regarding education at both the extensive and intensive margins. The results of this paper demonstrate that in the presence of robust social protection programs encouraging grade-school attendance, shocks to a low to mid-skilled tradable service sector shape enrollment decisions at the juncture between secondary and post-secondary education. When there is sizable support for schooling expenses from the government, the primary income-related effect is reduced illness-driven absenteeism likely due to higher medical expenditures. Such supports for tertiary education may be necessary to prevent a “Secondary Schooling-Trap" from occurring in countries like Jamaica attempting to make the final leap to high-income status.

For policymakers considering tourism specialization as a development strategy, the results on local service provision underscore the need to consider how growth of sectors requiring in-person consumption can produce positive spillovers for some while creating challenges for others. For researchers, a productive line of inquiry would examine the distributional impacts of different service-sector categories on human capital accumulation, with attention to how crowding out of local services may constrain educational access through effects on transportation costs or other goods and services. Such work could illuminate a tension between the income effects of local service-sector expansion and localized inflation.

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7 Appendix

7.1 Additional Main Specification Results

Table 7: Skills Program Enrollment and Completion, IV Estimates

Dependent Variables:	Enrolled in skills program		Skills program diploma
Model:	(1)	(2)	(3)
Variables			
Tourism exp., lag 1 (100M USD)	-0.006 (0.010)	-0.006 (0.010)	0.040 (0.039)
Fit statistics			
Bartik F-stat (1st stage)	114.72	114.72	189.77
Observations	1,576	1,576	1,866
R^2	0.14540	0.14540	0.14185
Mean of dep. var.	0.003	0.003	0.077

Notes: Sample: children ages 5–19. Controls for gender, age, and enrolled household siblings are included in all specifications but not shown. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 8: School Attendance, IV Estimates

Dependent Variables:	Days sent per week		Kept from school (any)
Model:	(1)	(2)	(3)
Variables			
Tourism exp., lag 1 (100M USD)	-0.245 (0.721)	-0.245 (0.721)	0.043 (0.036)
Fit statistics			
Bartik F-stat (1st stage)	862.81	862.81	863.27
Observations	11,412	11,412	11,881
R^2	0.03435	0.03435	0.22997
Mean of dep. var.	18.803	18.803	0.139

Notes: Sample: currently enrolled children ages 5–19. Controls for gender, age, and enrolled household siblings are included in all specifications but not shown. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 9: School Expenditure by Category (Log), IV Estimates

Dependent Variables:	Log total school spending	Log tuition	Log books	Log transport	Log uniform	Log lunch and snacks
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Tourism exp., lag 1 (100M USD)	0.648 (0.471)	-0.596 (1.95)	-4.01** (1.98)	2.29 (2.18)	-1.27 (0.960)	1.32 (1.05)
Fit statistics						
Bartik F-stat (1st stage)	633.32	431.64	634.72	638.01	657.07	632.38
Observations	10,352	9,246	10,281	10,283	10,304	10,337
R^2	0.02600	0.08843	0.06796	0.13568	0.02279	0.02790
Mean of dep. var.	6.347	-17.676	-0.619	-2.766	2.586	4.455

Notes: Sample: enrolled children; 2014 excluded due to survey design. Controls for gender, age, and enrolled household siblings are included in all specifications but not shown. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 10: Textbook Access, IV Estimates

Dependent Variable:	Textbook access (1–3)			
Model:	(1)	(2)	(3)	(4)
Variables				
Tourism exp.,	0.052	0.023	0.245	-0.071
lag 1 (100M USD)	(0.104)	(0.109)	(0.185)	(0.090)
Fit statistics				
Bartik F-stat (1st stage)	809.04	619.87	173.60	2,791.1
Observations	11,303	10,271	4,234	7,069
R^2	0.07577	0.08069	0.02967	0.09398
Mean of dep. var.	1.348	1.342	1.326	1.361

Notes: Sample: enrolled children ages 5–19. Outcome is ordinal: 1 = has all required textbooks, 2 = has some, 3 = has none. Harmonized across a questionnaire renumbering in 2015 and a question redesign in 2018 (required and supplemental books). Controls for gender, age, and enrolled household siblings are included in all specifications but not shown. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 11: Reasons for Lacking Textbooks: Availability and Other, IV Estimates

Dependent Variables:	School has no books	Can't find books	Books not necessary	Other reason
Model:	(1)	(2)	(3)	(4)
Variables				
Tourism exp.,	0.111	0.045*	-0.017	0.023
lag 1 (100M USD)	(0.070)	(0.026)	(0.043)	(0.071)
Fit statistics				
Bartik F-stat (1st stage)	205.75	205.75	205.75	205.75
Observations	3,878	3,878	3,878	3,878
R^2	0.78055	0.87009	0.83000	0.81950
Mean of dep. var.	0.192	0.175	0.183	0.185

Notes: Sample: enrolled children ages 5–19 who report lacking at least some required textbooks. Each column is a binary outcome (0/1) for one reason. Pre-2012 years derived from a single-response question; 2012–2019 from multi-select binary columns. Controls for gender, age, and enrolled household siblings are included in all specifications but not shown. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 12: School Expenditure, Supplementary Categories (Levels), IV Estimates

Dependent Variables:	Exam fees (USD)	Other fees (USD)	Extra lessons (USD)	Other supplies (USD)	Boarding (USD)
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp.,	-3.25	18.5	4.12	3.28	2.06
lag 1 (100M USD)	(5.64)	(31.6)	(18.0)	(9.95)	(2.87)
Fit statistics					
Bartik F-stat (1st stage)	718.06	844.15	661.56	618.80	819.78
Observations	10,186	6,692	10,209	10,287	10,156
R^2	0.10127	0.07354	0.05071	0.09850	0.00741
Mean of dep. var.	8.476	30.524	33.619	35.548	1.054

Notes: Sample: enrolled children; real 2024 USD; 2014 excluded due to survey design. Other fees not available for 2017 (questionnaire split the category into two sub-components that year); 2017 is not in the analysis sample. Controls for gender, age, and enrolled household siblings are included in all specifications but not shown. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 13: Household Electronics Spending (Computers + Cellphones), IV Estimates

Dependent Variables:	Electronics (USD)	Log electronics spending	Computers (USD)	Log computer spending	Any computer spending (0/1)
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp.,	-205.1	0.500	-33.4	0.345	0.012
lag 1 (100M USD)	(323.8)	(2.14)	(64.8)	(0.400)	(0.014)
HHSIZE2	48.1***	0.361***	7.58***	0.064***	0.002***
	(5.02)	(0.040)	(1.51)	(0.013)	(0.0005)
RURAL	-2.09	-0.175	0.837	-0.107	-0.004
	(28.1)	(0.165)	(10.6)	(0.066)	(0.002)
Fit statistics					
Bartik F-stat (1st stage)	451.13	451.13	451.13	451.13	451.13
Observations	23,823	23,823	23,823	23,823	23,823
R^2	0.01943	0.07642	0.00607	0.01132	0.01133
Mean of dep. var.	293.977	-16.379	40.401	-20.370	0.013

Notes: One observation per household per year. Real 2024 USD. *Electronics* = annual household spending on information processing equipment (ITEM_CD 3202: computers, tablets) plus cellphones (ITEM_CD 3481). *Computers* = information processing equipment only (ITEM_CD 3202). Controls: household size and urban/rural indicator. Development area and year fixed effects included; standard errors clustered by development area.

Table 14: Book Spending: Intensive vs. Extensive Margin, IV Estimates

Dependent Variables:	Books (USD)	Log books	Any book spending (0/1)	Books (USD)	Log books
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp.,	-24.5**	-4.01**	-0.157*	-10.9	-0.097
lag 1 (100M USD)	(9.83)	(1.98)	(0.084)	(12.1)	(0.269)
Fit statistics					
Bartik F-stat (1st stage)	634.72	634.72	634.72	408.67	408.67
Observations	10,281	10,281	10,281	8,339	8,339
R^2	0.08583	0.06796	0.06817	0.12519	0.09772
Mean of dep. var.	70.385	-0.619	0.811	86.776	4.062

Notes: Sample: enrolled children; real 2024 USD; 2014 excluded due to survey design. Columns (1)–(2) use the full enrolled sample. *Any books (0/1)* is a linear probability model (LPM) for the extensive margin; the coefficient is the percentage-point change in the probability of positive book spending. Columns (4)–(5) restrict to households with positive book spending (*buyers only*), isolating the intensive margin. If columns (1)–(2) are negative but (4)–(5) are near zero, the effect operates entirely through the extensive margin. The buyers-only sample also serves as a robustness check against potential coding inconsistencies in which zero spending is recorded as `NA` in some survey years. Controls for gender, age, and enrolled household siblings are included in all specifications but not shown. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 15: School Expenditure, Supplementary Categories (Log), IV Estimates

Dependent Variables:	Log exam fees	Log other fees	Log extra lessons	Log other supplies	Log boarding
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp.,	-0.176	3.16	0.969	0.005	0.054
lag 1 (100M USD)	(0.820)	(5.69)	(0.866)	(3.25)	(0.066)
Fit statistics					
Bartik F-stat (1st stage)	718.06	844.15	661.56	618.80	819.78
Observations	10,186	6,692	10,209	10,287	10,156
R^2	0.13206	0.15484	0.04802	0.04221	0.00827
Mean of dep. var.	-19.426	-10.305	-17.402	-0.228	-20.669

Notes: Sample: enrolled children; 2014 excluded due to survey design. Other fees not available for 2017 (questionnaire split into two sub-components that year); 2017 is not in the analysis sample. Controls for gender, age, and enrolled household siblings are included in all specifications but not shown. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 16: Distance to School, IV Estimates

Dependent Variables:	Dist. to school (km)	Dist. nearest primary (km)	Dist. nearest secondary (km)
Model:	(1)	(2)	(3)
Variables			
Tourism exp.,	-0.460	0.755	-1.26
lag 1 (100M USD)	(1.70)	(0.720)	(1.22)
Fit statistics			
Bartik F-stat (1st stage)	788.95	764.25	763.70
Observations	11,072	11,031	11,057
R^2	0.17412	0.04330	0.21780
Mean of dep. var.	8.825	2.355	5.893

Notes: Sample: enrolled children ages 5–19. Distances converted to kilometres from reported value and unit (miles, km, yards, metres, or chains; 1 chain = 20.1168 m). Values above 150 km treated as data-entry errors. Controls for gender, age, and enrolled household siblings are included in all specifications but not shown. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 17: Mode of Transport to School, IV Estimates

Dependent Variables:	Public transport	Walk	Private vehicle	School bus	Other transport
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp.,	0.043	-0.049	0.021	-0.016	0.001
lag 1 (100M USD)	(0.064)	(0.089)	(0.047)	(0.018)	(0.008)
Fit statistics					
Bartik F-stat (1st stage)	1,000.7	1,000.7	1,000.7	1,000.7	1,000.7
Observations	11,184	11,184	11,184	11,184	11,184
R^2	0.12810	0.13050	0.06501	0.02796	0.01205
Mean of dep. var.	0.664	0.269	0.051	0.009	0.006

Notes: Sample: enrolled children ages 5–19. Each column is a binary outcome (0/1) for one mode: 1 = public transport, 2 = walk, 3 = private vehicle, 4 = school bus, 5 = other. Controls for gender, age, and enrolled household siblings are included in all specifications but not shown. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 18: Parental Share of School Expenses, IV Estimates

Dependent Variables:	Parent pays: none (0%)	Parent pays: 25%	Parent pays: half (50%)	Parent pays: 75%	Parent pays: all (100%)
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp.,	-0.065	0.006	0.035^{**}	-0.011^{**}	0.032
lag 1 (100M USD)	(0.051)	(0.009)	(0.016)	(0.005)	(0.045)
Fit statistics					
Bartik F-stat (1st stage)	863.27	863.27	863.27	863.27	863.27
Observations	11,881	11,881	11,881	11,881	11,881
R^2	0.57993	0.02545	0.08954	0.00361	0.27558
Mean of dep. var.	0.201	0.004	0.020	0.003	0.155

Notes: Sample: enrolled children ages 5–19. Each column is a binary indicator equal to 1 if the parent or guardian pays exactly that share of total school expenses. Outcome derived from a harmonized 5-level ordinal (0/25/50/75/100%). The 0% category is unavailable in string-label survey years (2005–2007, 2010, 2016). Controls for gender, age, and enrolled household siblings are included in all specifications but not shown. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 19: MOEYC Share of School Expenses, IV Estimates

Dependent Variables:	MOEYC pays: none (0%)	MOEYC pays: 25–50%	MOEYC pays: 75–100%	MOEYC pays: all (100%)
Model:	(1)	(2)	(3)	(4)
Variables				
Tourism exp.,	0.060	-0.058	0.046	0.051
lag 1 (100M USD)	(0.146)	(0.066)	(0.112)	(0.109)
Fit statistics				
Bartik F-stat (1st stage)	764.30	764.30	764.30	764.30
Observations	8,161	8,161	8,161	8,161
R^2	0.39459	0.11236	0.13891	0.13863
Mean of dep. var.	0.477	0.026	0.057	0.053

Notes: Sample: enrolled children ages 5–19, years 2006–2014 (MOEYC share column absent in 2018–2019). MOEYC = Ministry of Education, Youth and Culture. Each column is a binary indicator for the share of school expenses paid by MOEYC. Controls for gender, age, and enrolled household siblings are included in all specifications but not shown. All specifications include development area and year fixed effects; standard errors clustered by development area.

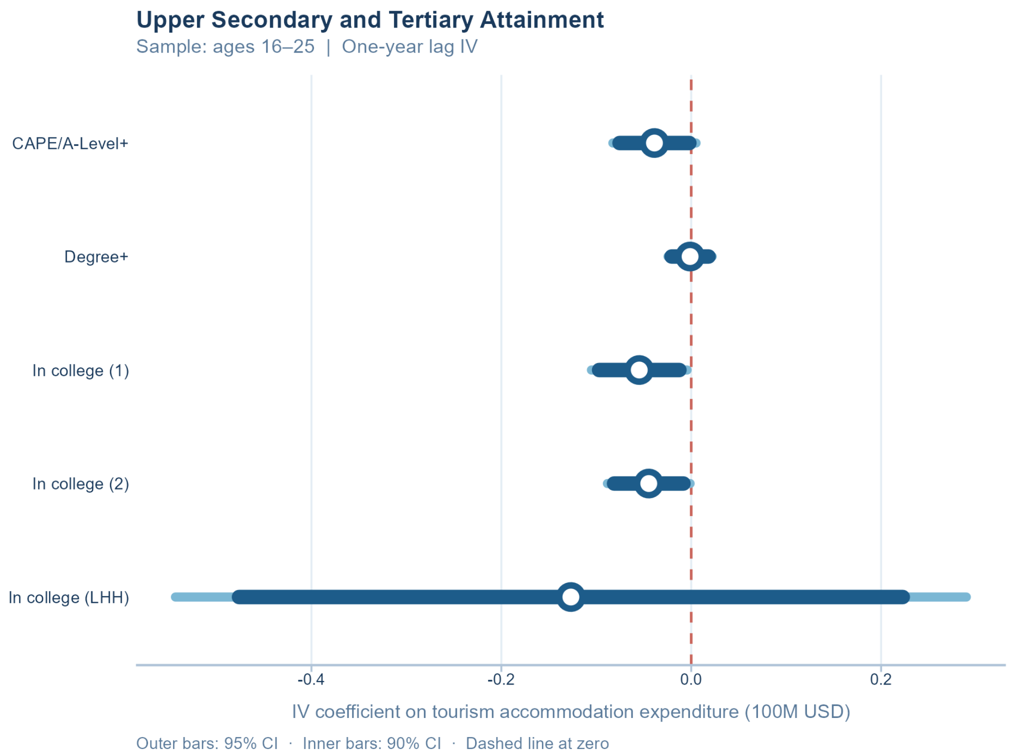

Figure 5: Upper Secondary and Tertiary Attainment: IV Coefficient Estimates Tourism expenditure coefficients (one-year lag) from the IV regressions in Table ??. Wide bars show 95% confidence intervals; narrow bars show 90% intervals. Dashed line marks zero.

7.2 Instrument Summary Statistics and First Stage Regressions

[Appendix placeholder.]

7.3 Shift-Share Instrument Diagnostics and Summary Statistics

[Appendix placeholder.]

7.4 Schooling Expenditures Additional Analysis

I next test whether the negative coefficient reflects extensive margin shifts, with some households shifting to spending no money on textbooks, or intensive margin effects of households spending less on textbooks than they were before the shock. The coefficient estimate is significant at the 10 percent level, and indicates that 100 million U.S. Dollars of additional

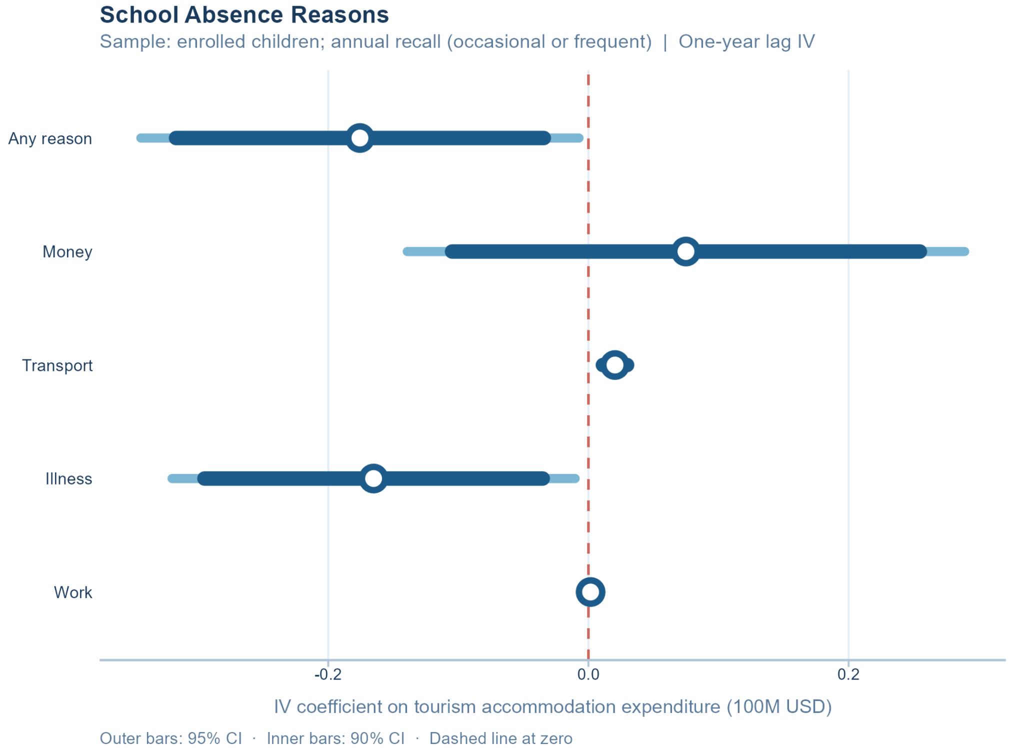

Figure 6: Absence Reasons: IV Coefficient Estimates
Tourism expenditure coefficients (one-year lag) from the IV regressions in Table 4. Wide bars show 95% confidence intervals; narrow bars show 90% intervals. Dashed line marks zero.

revenue produce a 15.7 percentage point decline in the share of students whose households spend any money on books, a 19 percent decline.

Table 20: Transport Absence by Urban/Rural, IV Estimates

Dependent Variable:		Kept out, transport	
Model:	(1)	(2)	(3)
Variables			
Tourism exp., lag 1 (100M USD)	0.020***	0.011	0.025***
	(0.006)	(0.008)	(0.009)
Fit statistics			
Bartik F-stat (1st stage)	863.27	190.86	2,864.2
Observations	11,881	4,501	7,380
R^2	0.00211	-0.00407	0.02772
Mean of dep. var.	0.003	0.002	0.004

Notes: Outcome is an indicator equal to 1 if the child was kept from school occasionally or frequently due to transportation costs in the past year. Columns (2) and (3) restrict the sample to urban and rural enrolled children, respectively. Controls for gender, age, and enrolled household siblings are included in all specifications but not shown. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 21: Textbook Access by Category, IV Estimates

Dependent Variables:	Has all textbooks	Has some textbooks	Has no textbooks
Model:	(1)	(2)	(3)
Variables			
Tourism exp., lag 1 (100M USD)	-0.043	0.034	0.009
	(0.076)	(0.056)	(0.037)
Fit statistics			
Bartik F-stat (1st stage)	809.04	809.04	809.04
Observations	11,303	11,303	11,303
R^2	0.09628	0.09652	0.03105
Mean of dep. var.	0.708	0.236	0.056

Notes: Sample: enrolled children ages 5–19. Each column is a binary outcome (0/1). *Has all*: child has all required textbooks. *Has some*: child has a partial set. *Has none*: child has no required textbooks. Harmonized across a questionnaire change in 2015 (question renumbered) and 2018 (required and supplemental books combined). Controls for gender, age, and enrolled household siblings are included in all specifications but not shown. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 22: Reasons for Lacking Textbooks: Financial Barriers, IV Estimates

Dependent Variables:	Unpaid school fees	Unpaid book fees	Money problems	Books too expensive
Model:	(1)	(2)	(3)	(4)
Variables				
Tourism exp., lag 1 (100M USD)	0.013	0.0006	-0.025	0.246**
	(0.048)	(0.046)	(0.099)	(0.118)
Fit statistics				
Bartik F-stat (1st stage)	205.75	205.75	205.75	205.75
Observations	3,878	3,878	3,878	3,878
R^2	0.92565	0.93985	0.09238	0.50190
Mean of dep. var.	0.166	0.164	0.860	0.306

Notes: Sample: enrolled children ages 5–19 who report lacking at least some required textbooks. Each column is a binary outcome (0/1) for one reason. Pre-2012 years derived from a single-response question (at most one reason equals 1 per observation); 2012–2019 from multi-select binary columns. Controls for gender, age, and enrolled household siblings are included in all specifications but not shown. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 23: Household Socioeconomic Status by Dependent Count

No. dependents	*N* households	*Poverty rate (%)*	*Pop. decile*	*Per-cap. exp. (USD)*
1	2,367	10.8%	6.4	3,763
2	1,851	13.5%	5.4	2,983
3	975	25.6%	4.4	2,421
4	483	32.4%	3.5	1,945
5	197	39.3%	3.3	1,777
6+	166	57.0%	2.3	1,378

Notes: One observation per household-year, pooled across JSLC regression years. *Poverty rate* is the JSLC consumption poverty headcount indicator. *Pop. decile* is the household's position in the national per-capita expenditure distribution (1 = lowest decile, 10 = highest). *Per-cap. exp.* is annual per-capita household expenditure in real 2024 USD. Households with zero dependents aged 19 or under are excluded.

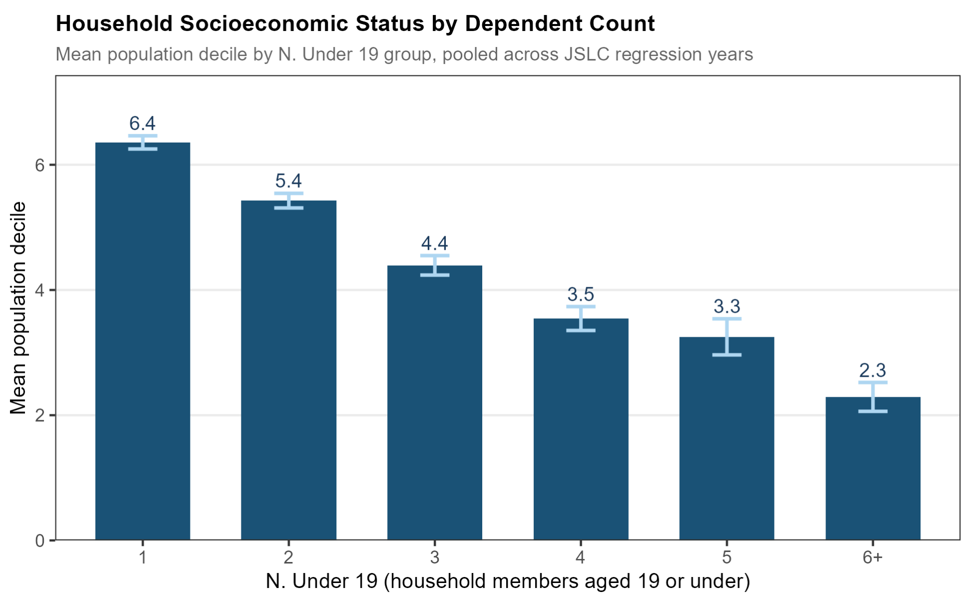

Figure 7: Mean Population Decile by Number of Household Dependents
Mean population decile (1 = lowest, 10 = highest) by number of household members aged 19 or under, pooled across JSLC regression years. Households with more dependents occupy systematically lower deciles, supporting the use of N. Under 19 as an inverse proxy for household socioeconomic status.

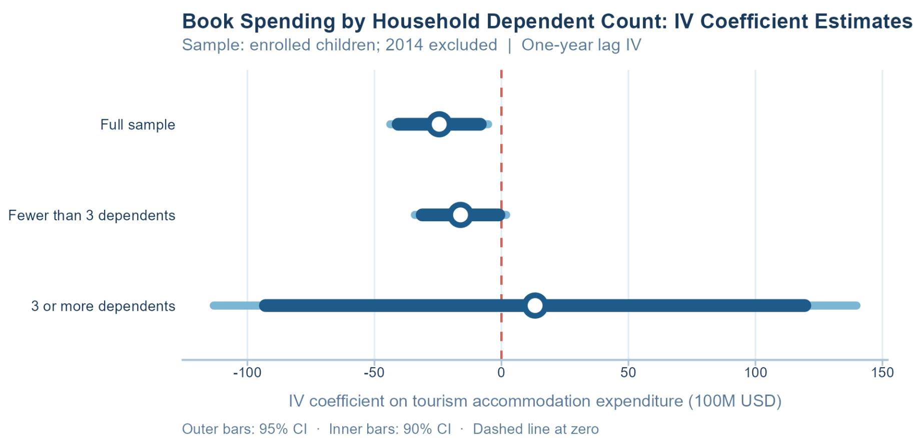

Figure 8: Book Spending by Household Dependent Count: IV Coefficient Estimates
Tourism expenditure coefficients (one-year lag) on book spending (real 2024 USD) for the full sample and for subsamples split by N. Under 19. Wide bars show 95% confidence intervals; narrow bars show 90% intervals. Dashed line marks zero.

7.5 Additional Regression Results: Contemporaneous Specification

Tables 24–30 replicate the main results using a contemporaneous specification in which development area tourism expenditures and the Bartik instrument enter without a lag.

Table 24: School Enrollment, IV Estimates (Contemporaneous)

Dependent Variable:	Enrolled in school	
Model:	(1)	(2)
Variables		
Tourism exp.	-0.247	-0.247
(100M USD)	(0.404)	(0.404)
Female	0.006*	0.006*
	(0.003)	(0.003)
Age	-0.038***	-0.038***
	(0.0010)	(0.0010)
No. enrolled siblings (HH)	0.002	0.002
	(0.004)	(0.004)
Fit statistics		
Bartik F-stat (1st stage)	34.270	34.270
Observations	13,687	13,687
R^2	0.17776	0.17776
Mean of dep. var.	0.879	0.879

Notes: Sample: children ages 5–19. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 25: Dropout Reasons, IV Estimates (Contemporaneous)

Dependent Variables:	Dropout: financial	Dropout: family	Dropout: no interest	Dropout: pregnancy
Model:	(1)	(2)	(3)	(4)
Variables				
Tourism exp.	-0.386	-0.070	-0.004	0.315
(100M USD)	(0.361)	(0.154)	(0.184)	(0.310)
Female	-0.034***	-0.005	-0.041***	0.081***
	(0.013)	(0.009)	(0.009)	(0.011)
Age	-0.002	-0.004**	-0.005	-0.004*
	(0.003)	(0.002)	(0.003)	(0.002)
No. enrolled siblings (HH)	0.006	-0.004	-0.005	0.011*
	(0.006)	(0.003)	(0.003)	(0.006)
Fit statistics				
Bartik F-stat (1st stage)	5.6526	5.6526	5.6526	5.6526
Observations	1,655	1,655	1,655	1,655
R^2	-0.37923	0.00446	0.04463	-0.20542
Mean of dep. var.	0.039	0.020	0.044	0.044

Notes: Sample: children ages 5–19 not currently enrolled. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 26: School Attendance, IV Estimates (Contemporaneous)

Dependent Variables:	Days sent per week		Kept from school (any)
Model:	(1)	(2)	(3)
Variables			
Tourism exp.	1.03	1.03	0.097
(100M USD)	(4.16)	(4.16)	(0.365)
Female	0.052	0.052	0.001
	(0.113)	(0.113)	(0.005)
Age	-0.026**	-0.026**	-0.007***
	(0.011)	(0.011)	(0.0010)
No. enrolled siblings (HH)	-0.044	-0.044	-0.007
	(0.079)	(0.079)	(0.006)
Fit statistics			
Bartik F-stat (1st stage)	27.989	27.989	29.229
Observations	11,557	11,557	12,032
R^2	0.02671	0.02671	0.22087
Mean of dep. var.	18.804	18.804	0.138

Notes: Sample: currently enrolled children ages 5–19. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 27: School Absence Reasons, IV Estimates (Contemporaneous)

Dependent Variables:	Kept out, any reason	Kept out, money	Kept out, work	Kept out, illness	Kept out, transport
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp.	-0.314	-0.124	-0.004	-0.298	0.085
(100M USD)	(0.371)	(0.352)	(0.014)	(0.247)	(0.088)
Female	-0.027***	-0.018***	-0.0004	-0.002	-0.0001
	(0.007)	(0.007)	(0.0004)	(0.004)	(0.001)
Age	-0.002	0.002	3.07×10^{-5}	-0.004***	0.0005***
	(0.001)	(0.001)	(5.68×10^{-5})	(0.0005)	(0.0002)
No. enrolled siblings (HH)	0.019***	0.023***	-5.11×10^{-5}	0.0003	-0.0004
	(0.006)	(0.006)	(0.0002)	(0.003)	(0.0010)
Fit statistics					
Bartik F-stat (1st stage)	29.229	29.229	29.229	29.229	29.229
Observations	12,032	12,032	12,032	12,032	12,032
R^2	0.04245	0.04741	0.00062	-0.14018	-0.31279
Mean of dep. var.	0.190	0.108	0.000	0.063	0.003

Notes: Sample: enrolled children; absence measured by annual recall (occasional or frequent). All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 28: Transport Absence by Urban/Rural, IV Estimates (Contemporaneous)

Dependent Variable:	Kept out, transport		
Model:	(1)	(2)	(3)
Variables			
Tourism exp.	0.085	0.023	0.224
(100M USD)	(0.088)	(0.023)	(0.387)
Female	-0.0001	-0.002	-0.0001
	(0.001)	(0.002)	(0.002)
Age	0.0005***	0.0005**	0.0004*
	(0.0002)	(0.0002)	(0.0002)
No. enrolled siblings (HH)	-0.0004	0.0010	-0.002
	(0.0010)	(0.001)	(0.003)
Fit statistics			
Bartik F-stat (1st stage)	29.229	22.257	15.899
Observations	12,032	4,501	7,531
R^2	-0.31279	-0.06044	-0.48178
Mean of dep. var.	0.003	0.002	0.004

Notes: Outcome is an indicator equal to 1 if the child was kept from school occasionally or frequently due to transportation costs in the past year. Columns (2) and (3) restrict the sample to urban and rural enrolled children, respectively. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 29: School Expenditure by Category, IV Estimates (Contemporaneous)

Dependent Variables:	Tuition (USD)	Books (USD)	Transport (USD)	Uniform (USD)	Lunch and snacks (USD)
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp.	-72.1	-157.4	52.6	-133.7	-71.1
(100M USD)	(113.5)	(181.5)	(146.5)	(142.7)	(235.3)
Female	-3.30	6.32***	24.9***	-7.42***	15.5**
	(6.17)	(1.75)	(7.92)	(1.78)	(6.82)
Age	-7.25***	1.03***	31.5***	2.61***	29.1***
	(0.800)	(0.247)	(2.51)	(0.265)	(1.24)
No. enrolled siblings (HH)	-11.7***	-7.50***	-12.3***	-6.69***	-26.7***
	(2.69)	(1.99)	(3.14)	(1.49)	(3.99)
Fit statistics					
Bartik F-stat (1st stage)	28.389	18.818	21.921	21.131	19.856
Observations	9,384	10,427	10,429	10,448	10,483
R^2	0.03084	-0.54049	0.07799	-0.18035	0.12014
Mean of dep. var.	47.787	70.273	257.486	88.431	463.130

Notes: Sample: enrolled children; real 2024 USD; 2014 excluded due to survey design. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 30: School Expenditure by Category (Log), IV Estimates (Contemporaneous)

Dependent Variables:	Log tuition	Log books	Log transport	Log uniform	Log lunch and snacks
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp.	-5.52	-8.51	6.60	-0.261	-0.244
(100M USD)	(12.0)	(11.6)	(7.23)	(2.90)	(3.74)
Female	0.049	0.695***	0.901***	0.085	-0.070
	(0.194)	(0.193)	(0.231)	(0.138)	(0.111)
Age	-0.550***	-0.334***	1.11***	0.020	0.179***
	(0.034)	(0.037)	(0.041)	(0.020)	(0.019)
No. enrolled siblings (HH)	-0.324*	-0.796***	-0.438***	-0.232**	-0.192***
	(0.162)	(0.130)	(0.129)	(0.102)	(0.060)
Fit statistics					
Bartik F-stat (1st stage)	28.389	18.818	21.921	21.131	19.856
Observations	9,384	10,427	10,429	10,448	10,483
R^2	0.01559	-0.02140	0.08859	0.02339	0.03028
Mean of dep. var.	-17.707	-0.628	-2.791	2.584	4.454

Notes: Sample: enrolled children; 2014 excluded due to survey design. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 31: Examination Attainment, IV Estimates (Contemporaneous)

Dependent Variables:	GNAT	CXC Basic	CXC General	CXC: math and English	CAPE/ A-Level	Degree
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Tourism exp.	0.097	0.082	0.051	0.005	-0.234	0.014
(100M USD)	(0.174)	(0.155)	(0.110)	(0.066)	(0.204)	(0.024)
Female	0.032***	0.031***	0.026***	0.011***	0.015***	0.004
	(0.004)	(0.004)	(0.004)	(0.002)	(0.006)	(0.003)
Age	0.021***	0.019***	0.016***	0.007***	0.010***	0.003***
	(0.0008)	(0.0009)	(0.0009)	(0.0008)	(0.001)	(0.0006)
No. enrolled siblings (HH)	-0.006**	-0.006**	-0.006**	-0.002**	-0.006**	-0.003**
	(0.003)	(0.003)	(0.003)	(0.0009)	(0.002)	(0.001)
Fit statistics						
Bartik F-stat (1st stage)	34.270	34.270	34.270	34.270	23.957	27.499
Observations	13,687	13,687	13,687	13,687	6,275	6,045
R^2	0.11670	0.11635	0.10554	0.05221	-0.14401	0.01764
Sample						
School-age (5–19)	Yes	Yes	Yes	Yes	No	No
A-Level age (16–25)	No	No	No	No	Yes	No
College-age (17–27)	No	No	No	No	No	Yes
Mean of dep. var.	0.061	0.056	0.046	0.019	0.041	0.009

Notes: Sample varies by outcome: ages 5–19 for GNAT through CXC qualifications; ages 16–25 for A-Level and Degree. Each outcome is a binary indicator equal to 1 if the respondent has attained at least that qualification level (or higher). All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 32: Upper Secondary and Tertiary Attainment, IV Estimates (Contemporaneous)

Dependent Variables:	At least CAPE/A-Level	At least degree		Currently in college	
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp. (100M USD)	-0.182	0.008	-0.092	-0.123	0.108
	(0.178)	(0.022)	(0.107)	(0.142)	(0.227)
Female	0.029***	0.009***	0.036***	0.036***	0.032*
	(0.005)	(0.002)	(0.006)	(0.007)	(0.017)
Age	0.010***	0.004***	-0.0001	0.001	-0.0005
	(0.002)	(0.0008)	(0.0008)	(0.001)	(0.002)
No. enrolled siblings (HH)				-0.005*	
				(0.003)	
Fit statistics					
Bartik F-stat (1st stage)	32.058	32.058	32.058	23.957	5.8504
Observations	7,879	7,879	7,879	6,275	315
R^2	-0.04551	0.02229	0.01975	0.01054	0.26013
Mean of dep. var.	0.049	0.010	0.067	0.065	0.029

Notes: Sample: ages 16–25. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 33: Skills Program Enrollment and Completion, IV Estimates (Contemporaneous)

Dependent Variables:	Enrolled in skills program		Skills program diploma
Model:	(1)	(2)	(3)
Variables			
Tourism exp.	-0.013	-0.013	0.125
(100M USD)	(0.041)	(0.041)	(0.174)
Female	0.003	0.003	-0.003
	(0.002)	(0.002)	(0.011)
Age	-0.0006	-0.0006	0.014***
	(0.0006)	(0.0006)	(0.004)
No. enrolled siblings (HH)	-0.001*	-0.001*	-0.009*
	(0.0006)	(0.0006)	(0.005)
Fit statistics			
Bartik F-stat (1st stage)	3.5134	3.5134	5.6541
Observations	1,594	1,594	1,886
R^2	0.13597	0.13597	0.12238
Mean of dep. var.	0.003	0.003	0.076

Notes: Sample: children ages 5–19. All specifications include development area and year fixed effects; standard errors clustered by development area.

7.6 Additional Regression Results: Two-Year Lagged Specification

Tables 34–40 replicate the main results using a two-year lag specification in which development area tourism expenditures and the Bartik instrument enter with a two-year lag.

Table 34: School Enrollment, IV Estimates (Lag 2)

Dependent Variable:	Enrolled in school	
Model:	(1)	(2)
Variables		
Tourism exp., lag 2 (100M USD)	-0.044	-0.044
	(0.071)	(0.071)
Female	0.006*	0.006*
	(0.003)	(0.003)
Age	-0.038***	-0.038***
	(0.001)	(0.001)
No. enrolled siblings (HH)	-0.0001	-0.0001
	(0.002)	(0.002)
Fit statistics		
Bartik F-stat (1st stage)	502.45	502.45
Observations	13,518	13,518
R^2	0.25499	0.25499
Mean of dep. var.	0.879	0.879

Notes: Sample: children ages 5–19. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 35: Dropout Reasons, IV Estimates (Lag 2)

Dependent Variables:	Dropout: financial	Dropout: family	Dropout: no interest	Dropout: pregnancy
Model:	(1)	(2)	(3)	(4)
Variables				
Tourism exp., lag 2 (100M USD)	0.019	0.074	0.137	-0.102
	(0.050)	(0.067)	(0.151)	(0.099)
Female	-0.031***	-0.004	-0.039***	0.079***
	(0.011)	(0.009)	(0.010)	(0.011)
Age	-0.003	-0.005**	-0.005	-0.003
	(0.004)	(0.002)	(0.003)	(0.002)
No. enrolled siblings (HH)	0.008	-0.003	-0.003	0.009*
	(0.006)	(0.003)	(0.004)	(0.005)
Fit statistics				
Bartik F-stat (1st stage)	67.093	67.093	67.093	67.093
Observations	1,637	1,637	1,637	1,637
R^2	0.08795	0.03729	0.01672	0.05894
Mean of dep. var.	0.038	0.020	0.043	0.044

Notes: Sample: children ages 5–19 not currently enrolled. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 36: School Attendance, IV Estimates (Lag 2)

Dependent Variables:	Days sent per week		Kept from school (any)
Model:	(1)	(2)	(3)
Variables			
Tourism exp.,	-0.107	-0.107	0.100*
lag 2 (100M USD)	(1.51)	(1.51)	(0.058)
Female	0.035	0.035	0.0008
	(0.113)	(0.113)	(0.005)
Age	-0.029***	-0.029***	-0.007***
	(0.011)	(0.011)	(0.0008)
No. enrolled siblings (HH)	-0.035	-0.035	-0.006
	(0.078)	(0.078)	(0.004)
Fit statistics			
Bartik F-stat (1st stage)	434.51	434.51	439.36
Observations	11,412	11,412	11,881
R^2	0.03447	0.03447	0.22638
Mean of dep. var.	18.803	18.803	0.139

Notes: Sample: currently enrolled children ages 5–19. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 37: School Absence Reasons, IV Estimates (Lag 2)

Dependent Variables:	Kept out, any reason	Kept out, money	Kept out, work	Kept out, illness	Kept out, transport
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp.,	0.129	-0.154	-0.0008	0.369	-0.004
lag 2 (100M USD)	(0.211)	(0.105)	(0.002)	(0.319)	(0.021)
Female	-0.026***	-0.016**	-0.0004	-0.003	-5.49×10^{-5}
	(0.006)	(0.007)	(0.0004)	(0.006)	(0.001)
Age	-0.002	0.002	3.56×10^{-5}	-0.004***	0.0004***
	(0.001)	(0.0009)	(5.29×10^{-5})	(0.0006)	(0.0001)
No. enrolled siblings (HH)	0.015***	0.022***	-8.92×10^{-5}	-0.004	0.0005
	(0.004)	(0.005)	(0.0002)	(0.003)	(0.0006)
Fit statistics					
Bartik F-stat (1st stage)	439.36	439.36	439.36	439.36	439.36
Observations	11,881	11,881	11,881	11,881	11,881
R^2	0.13273	0.06072	0.00632	-0.04711	0.01883
Mean of dep. var.	0.188	0.108	0.001	0.063	0.003

Notes: Sample: enrolled children; absence measured by annual recall (occasional or frequent). All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 38: Transport Absence by Urban/Rural, IV Estimates (Lag 2)

Dependent Variable:	Kept out, transport		
Model:	(1)	(2)	(3)
Variables			
Tourism exp.,	-0.004	-6.38×10^{-5}	0.007
lag 2 (100M USD)	(0.021)	(0.019)	(0.009)
Female	-5.49×10^{-5}	-0.001	0.001
	(0.001)	(0.002)	(0.002)
Age	0.0004***	0.0005**	0.0003*
	(0.0001)	(0.0002)	(0.0002)
No. enrolled siblings (HH)	0.0005	0.001	5.08×10^{-5}
	(0.0006)	(0.001)	(0.0007)
Fit statistics			
Bartik F-stat (1st stage)	439.36	110.99	655.04
Observations	11,881	4,501	7,380
R^2	0.01883	0.01446	0.03019
Mean of dep. var.	0.003	0.002	0.004

Notes: Outcome is an indicator equal to 1 if the child was kept from school occasionally or frequently due to transportation costs in the past year. Columns (2) and (3) restrict the sample to urban and rural enrolled children, respectively. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 39: School Expenditure by Category, IV Estimates (Lag 2)

Dependent Variables:	Tuition (USD)	Books (USD)	Transport (USD)	Uniform (USD)	Lunch and snacks (USD)
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp.,	25.3	-9.77	-33.2	13.6	108.6
lag 2 (100M USD)	(40.0)	(11.2)	(68.3)	(23.2)	(89.4)
Female	-4.42	5.63***	25.7***	-8.40***	15.0**
	(6.76)	(1.87)	(7.99)	(1.77)	(6.54)
Age	-7.29***	1.15***	31.9***	2.69***	29.3***
	(0.770)	(0.247)	(2.48)	(0.262)	(1.30)
No. enrolled siblings (HH)	-12.8***	-9.34***	-11.7***	-8.39***	-28.2***
	(2.98)	(0.951)	(2.64)	(1.05)	(3.41)
Fit statistics					
Bartik F-stat (1st stage)	990.13	808.88	771.86	808.75	802.52
Observations	9,246	10,281	10,283	10,304	10,337
R^2	0.04441	0.09251	0.08191	0.05555	0.11900
Mean of dep. var.	48.388	70.385	258.767	88.556	464.417

Notes: Sample: enrolled children; real 2024 USD; 2014 excluded due to survey design. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 40: School Expenditure by Category (Log), IV Estimates (Lag 2)

Dependent Variables:	Log tuition	Any books expenses	Log transport	Log uniform	Log lunch and snacks
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp.,	-0.137			0.215	-0.842
lag 2 (100M USD)	(1.27)			(1.41)	(1.41)
Female	-0.010	0.024***	0.991***	0.053	-0.073
	(0.146)	(0.008)	(0.219)	(0.136)	(0.109)
Age	-0.552***	-0.014***	1.13***	0.025	0.183***
	(0.036)	(0.002)	(0.039)	(0.019)	(0.020)
No. enrolled siblings (HH)	-0.394***	-0.031***	-0.372***	-0.246**	-0.200***
	(0.084)	(0.005)	(0.105)	(0.098)	(0.058)
Tourism exp.,		-0.157*	2.29		
lag 1 (100M USD)		(0.084)	(2.18)		
Fit statistics					
Bartik F-stat (1st stage)	990.13			808.75	802.52
Bartik F-stat (1st stage)		634.72	638.01		
Observations	9,246	10,281	10,283	10,304	10,337
R^2	0.08912	0.06817	0.13568	0.02251	0.03080
Mean of dep. var.	-17.676	0.811	-2.766	2.586	4.455

Notes: Sample: enrolled children; 2014 excluded due to survey design. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 41: Examination Attainment, IV Estimates (Lag 2)

Dependent Variables:	GNAT	CXC Basic	CXC General	CXC: math and English	CAPE/ A-Level	Degree
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Tourism exp.,	0.0002	-0.0001	-0.021	-0.024	0.051	-0.022
lag 2 (100M USD)	(0.037)	(0.033)	(0.037)	(0.020)	(0.091)	(0.017)
Female	0.031***	0.030***	0.025***	0.011***	0.019***	0.004
	(0.004)	(0.004)	(0.004)	(0.002)	(0.004)	(0.003)
Age	0.021***	0.019***	0.016***	0.007***	0.010***	0.003***
	(0.0008)	(0.0009)	(0.0010)	(0.0008)	(0.001)	(0.0007)
No. enrolled siblings (HH)	-0.005**	-0.006**	-0.006**	-0.002**	-0.008***	-0.003***
	(0.002)	(0.002)	(0.002)	(0.0009)	(0.002)	(0.0010)
Fit statistics						
Bartik F-stat (1st stage)	502.45	502.45	502.45	502.45	229.12	273.84
Observations	13,518	13,518	13,518	13,518	6,209	5,985
R^2	0.14588	0.13853	0.11850	0.05207	0.03914	0.02594
Sample						
School-age (5–19)	Yes	Yes	Yes	Yes	No	No
A-Level age (16–25)	No	No	No	No	Yes	No
College-age (17–27)	No	No	No	No	No	Yes
Mean of dep. var.	0.061	0.056	0.046	0.020	0.041	0.009

Notes: Sample varies by outcome: ages 5–19 for GNAT through CXC qualifications; ages 16–25 for A-Level and Degree. Each outcome is a binary indicator equal to 1 if the respondent has attained at least that qualification level (or higher). All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 42: Upper Secondary and Tertiary Attainment, IV Estimates (Lag 2)

Dependent Variables:	At least CAPE/A-Level	At least degree		Currently in college	
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp., lag 2 (100M USD)	-0.029	-0.013	-0.044	-0.067	0.024
	(0.045)	(0.011)	(0.032)	(0.063)	(0.091)
Female	0.029***	0.009***	0.036***	0.038***	0.032*
	(0.005)	(0.002)	(0.006)	(0.006)	(0.018)
Age	0.010***	0.004***	-0.0004	0.001	-0.0003
	(0.002)	(0.0008)	(0.0008)	(0.0009)	(0.002)
No. enrolled siblings (HH)				-0.006**	
				(0.003)	
Fit statistics					
Bartik F-stat (1st stage)	296.30	296.30	296.30	229.12	10.237
Observations	7,799	7,799	7,799	6,209	304
R^2	0.05034	0.02775	0.03256	0.03247	0.26421
Mean of dep. var.	0.049	0.010	0.067	0.065	0.030

Notes: Sample: ages 16–25. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 43: Skills Program Enrollment and Completion, IV Estimates (Lag 2)

Dependent Variables:	Enrolled in skills program		Skills program diploma
Model:	(1)	(2)	(3)
Variables			
Tourism exp.,	-0.029	-0.029	-0.246
lag 2 (100M USD)	(0.120)	(0.120)	(0.302)
Female	0.003	0.003	-0.005
	(0.002)	(0.002)	(0.012)
Age	-0.0005	-0.0005	0.017***
	(0.001)	(0.001)	(0.004)
No. enrolled siblings (HH)	-0.001**	-0.001**	-0.009
	(0.0005)	(0.0005)	(0.005)
Fit statistics			
Bartik F-stat (1st stage)	9.2371	9.2371	14.643
Observations	1,576	1,576	1,866
R^2	0.13116	0.13116	0.11363
Mean of dep. var.	0.003	0.003	0.077

Notes: Sample: children ages 5–19. All specifications include development area and year fixed effects; standard errors clustered by development area.

7.7 Tourism Summary Statistics

[Appendix placeholder.]

7.8 Urban vs Rural School Distances

Table 44: School Distance by Urban/Rural Status

	Urban		*Rural*			
	N	Mean	N	Mean	Diff. (U−R)	
Distance to own school (km)	4,133	7.69	7,083	9.44	-1.75	***
Distance, nearest primary school (km)	4,140	1.90	7,036	2.62	-0.72	***
Distance, nearest secondary school (km)	4,142	3.01	7,060	7.55	-4.54	***

Notes: Sample: enrolled children ages 5–19, pooled across JSLC regression years. Distances converted to kilometres from reported value and unit (miles, km, yards, metres, or chains). Values above 150 km treated as data-entry errors and excluded. *Diff* = urban mean − rural mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (Welch two-sample t-test).

Table 45: Mode of Transport to School by Urban/Rural Status

	Urban (%)	*Rural* (%)	*Diff.* (pp)	*t-stat*	
Public transport	63.7%	67.9%	-4.2 pp	-4.52	***
Walk	26.0%	27.6%	-1.6 pp	-1.86	*
Private vehicle	8.2%	3.3%	4.9 pp	10.40	***
School bus	1.2%	0.8%	0.4 pp	2.11	**
Other	0.9%	0.5%	0.4 pp	2.62	***
Pooled sample shares: Public 66.4%, Walk 27%, Private 5.1%, School bus 0.9%, Other 0.6%					

Notes: Sample: enrolled children ages 5–19, pooled across JSLC regression years. Each row reports the share of children using that mode as their primary mode of transport to school. *Diff (pp)* = urban share − rural share in percentage points. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (Welch two-sample t-test on the binary mode indicator).

7.9 Additional Household and Education Summary Statistics

Table 46: School Enrollment Rates by Age Group and Year

	2006	2010	2012	2014	2018	2019	*Pooled*	*Change (pp)*
Enrollment rate (%)								
Overall (ages 5–27)	65.4%	67.6%	65.1%	64.3%	65.0%	63.7%	65.1%	—
Primary (5–11)	99.4%	99.3%	99.6%	99.4%	99.6%	99.4%	99.5%	—
Lower secondary (12–14)	99.0%	98.4%	99.3%	99.0%	98.7%	99.3%	99.0%	-0.5pp
Upper secondary (15–16)	87.7%	95.1%	93.6%	98.1%	96.7%	96.9%	94.4%	-4.7pp
Sixth form age (17–19)	35.8%	39.1%	43.1%	43.5%	52.9%	49.6%	44.9%	-49.5pp
Post-secondary (20–22)	6.6%	11.1%	12.8%	9.9%	13.5%	14.0%	12.0%	-32.9pp
Age 23–27	2.0%	5.2%	4.5%	8.2%	7.4%	6.5%	5.5%	-6.5pp

Notes: Entries show the share of individuals in the indicated age range enrolled in school in the given survey year. Regression years are 2006, 2010, 2012, 2014, 2018, and 2019. *Pooled* averages across all regression years. *Change (pp)* reports the percentage-point difference in the pooled enrollment rate relative to the previous age group, capturing the drop-off at each school-stage transition; "—" indicates no prior stage. Age groups correspond to Jamaican school stages: primary (grades 1–6, ages 5–11), lower secondary (grades 7–9, ages 12–14), upper secondary (grades 10–11, ages 15–16), sixth form (ages 17–19), and post-secondary young adults (ages 20–22 and 23–27). Sample: JSLC household members ages 5–27 in regression years.